

# A DICHOTOMY FOR ERDŐS PROBLEM 1112: EXISTENCE OF $r_k(d_1, d_2)$ FOR $k \geq 3$

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ABSTRACT. Let  $k \geq 3$  and  $1 \leq d_1 < d_2$  be integers. Erdős problem 1112 asks whether there is an integer  $r$  such that every lacunary sequence  $B = \{b_1 < b_2 < \dots\}$  of positive integers with  $b_{i+1} \geq r b_i$  admits a sequence  $A = \{a_1 < a_2 < \dots\}$  of positive integers with all gaps in  $[d_1, d_2]$  and  $(kA) \cap B = \emptyset$ , where  $kA$  is the  $k$ -fold sumset; write  $r_k(d_1, d_2)$  for the least such  $r$ .

We prove that  $r_k(d_1, d_2)$  exists if and only if  $d_2 \geq k + 1$ , with the explicit bound  $r_k(d_1, d_2) \leq 192 d_2$  when it exists. Below the threshold, non-existence holds in a strong form: for every prescribed sequence of pointwise lower bounds on the successive ratios of  $B$ , however rapidly growing, a single  $B$  meets the  $k$ -fold sumset of every admissible  $A$ .

The existence half uses a Beatty construction whose  $k$ -fold sumset threads the gaps a lacunary  $B$  must leave. The non-existence half reduces, through a word-combinatorial analysis of the gap sequence, to a bounded subset-sum theorem which we prove: every finite set  $G$  of at least three positive integers with  $\gcd(G) = 1$  and  $\max(G) = M$  admits a multiset of at most  $M - 1$  of its elements whose subset sums contain  $M$  consecutive integers. Its proof combines symbolic reductions with a finite certificate layer of a few hundred explicit instances, so the argument is computer-assisted. The dichotomy, including the certificate layer, has been formally verified in Lean 4, in the problem’s literal integer phrasing.

## 1. INTRODUCTION

1.1. **The problem.** Erdős problem 1112 [E1112] asks, verbatim:

Let  $1 \leq d_1 < d_2$  and  $k \geq 3$ . Does there exist an integer  $r$  such that if  $B = \{b_1 < \dots\}$  is a lacunary sequence of positive integers with  $b_{i+1} \geq r b_i$  then there exists a sequence of positive integers  $A = \{a_1 < \dots\}$  such that  $d_1 \leq a_{i+1} - a_i \leq d_2$  for all  $i \geq 1$  and  $(kA) \cap B = \emptyset$ , where  $kA$  is the  $k$ -fold sumset?

We pause over the phrase “does there exist an integer  $r$ ”. For a *negative*  $r$  the condition  $b_{i+1} \geq r b_i$  is vacuous, since  $B$  is positive; and  $r \leq 1$  never works, because then  $B = \{1, 2, 3, \dots\}$  is admissible and meets every nonempty  $kA$ . So the existence question over  $\mathbb{Z}$  is equivalent to the same question over the integers  $r \geq 2$ , and any witness is automatically at least 2. Accordingly we define

$r_k(d_1, d_2) :=$  the least integer  $r \geq 2$  with the stated property,

which is well defined precisely when the property holds for some  $r$ ; “ $r_k(d_1, d_2)$  exists” abbreviates that. (The formalization quantifies  $r$  over  $\mathbb{N}$ ; see §6.)

Throughout,  $A$  and  $B$  are infinite sets of positive integers, identified with their strictly increasing enumerations, and

$$kA = \{x_1 + \dots + x_k : x_1, \dots, x_k \in A\}$$

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Source, data and the Lean development: [https://github.com/beetree/math\\_erdos\\_1112](https://github.com/beetree/math_erdos_1112). See §9 for the release and reproduction instructions, and §7 for a statement of how this work was produced.

is the  $k$ -fold sumset *with repetitions allowed*. We call a sequence  $A$  with  $d_1 \leq a_{i+1} - a_i \leq d_2$  for all  $i$  a  $(d_1, d_2)$ -sequence, and we write “ $r_k(d_1, d_2)$  exists” for the existence of a ratio  $r$  as above. Explicitly, this means: there is an integer  $r$  such that *every* lacunary  $B$  of ratio  $r$  admits *some*  $(d_1, d_2)$ -sequence  $A$  with  $(kA) \cap B = \emptyset$ . The order of quantifiers matters:  $A$  is chosen after  $B$ .

*Remark 1.1* (Attribution). We solve the formulation recorded as Erdős problem 1112 on [erdosproblems.com](http://erdosproblems.com). As that page itself notes, the formulation is a generous interpretation of an ambiguous remark of Erdős and Graham [ErGr80, p. 18]; it may be more accurate to describe it as a problem *inspired by* them. Our results concern the formulation as stated there, which is the one quoted above.

**1.2. Prior results.** The problem page records the following. Erdős–Graham [ErGr80] and Bollobás–Hegvári–Jin [BHJ97] give  $r_2(2, 3) = 2$ ; Chen [Ch00] shows  $r_2(a, b) \leq 2$  for integers  $a < b$  with  $b \neq 2a$  (and  $r_2(a, 2a) \geq 2$ , nontrivial only in the real-ratio convention of that paper); and [BHJ97] shows that  $r_3(2, 3)$  does not exist, in a strong form. Tang and Yang [TaYa21] prove, for  $k \geq 3$ , that if the difference sequence of  $A$  is of *block type* then there is a  $B$  with  $(kA) \cap B \neq \emptyset$ . That restricted class is subsumed by our Lemma 3.5 (eventually periodic tails); the difficulty in the general problem lies precisely in the non-periodic (Sturmian) tails, which we handle in §3. The general existence question for  $k \geq 3$  is listed as open.

**1.3. Results.** Our main result answers the question completely.

*Theorem A* (The dichotomy). Let  $k \geq 3$  and  $1 \leq d_1 < d_2$ . Then:

- (1) (Existence.) If  $d_2 \geq k + 1$  then  $r_k(d_1, d_2)$  exists; explicitly,  $r_k(d_1, d_2) \leq 192 d_2$ .
- (2) (Non-existence, strong form.) If  $d_2 \leq k$  then  $r_k(d_1, d_2)$  does not exist. Indeed, for *every* prescribed sequence  $(r_i)_{i \geq 1}$  of nonnegative integers — with no monotonicity assumed — there is a *single*  $B$  with  $b_{i+1} \geq r_i b_i$  for all  $i$  such that  $(kA) \cap B \neq \emptyset$  for *every*  $(d_1, d_2)$ -sequence  $A$ . (Real lower bounds reduce to this: replace each  $r_i$  by  $\lceil r_i \rceil$ .)

Hence  $r_k(d_1, d_2)$  exists if and only if  $d_2 \geq k + 1$ , and the answer to the quoted question is: *yes precisely when  $d_2 \geq k + 1$* .

Note that  $d_1$  is irrelevant to the threshold. The constant 192 is not optimized, and we make no claim about the true order of  $r_k(d_1, d_2)$ . Part (2) is genuinely stronger than the negation of the question. It says that no prescribed sequence of *pointwise lower bounds on the successive ratios* of  $B$  — however rapidly growing, and fixed in advance — can rescue the existence of  $A$ . We state the scope exactly: the quantification is over ratio sequences  $(r_i)$ , not over arbitrary structural or arithmetic conditions that one might instead impose on  $B$ .

The consecutive-window dichotomy “ $r_k(d, d + 1)$  exists  $\iff d \geq k$ ” is exactly the  $d_2 = d + 1$  slice of Theorem A, for every  $k \geq 3$ .

*Remark 1.2* (The case  $k = 2$ , and why §3 needs  $k \geq 3$ ). Theorem A is stated for  $k \geq 3$ , as is the problem. For context,  $r_2(2, 3) = 2$  [ErGr80, BHJ97], and  $r_2(a, b) \leq 2$  for  $a < b$  with  $b \neq 2a$  [Ch00] (in the convention of [Ch00], which permits real ratios; under the integer convention of this paper a witness is automatically  $\geq 2$ ). The strong form of non-existence is *not* claimed at  $k = 2$ , and the hypothesis  $k \geq 3$  in §3 is not an artifact of the method: the width machinery genuinely fails there. There are balanced words whose two-index antidiagonal width  $W_2$  vanishes infinitely often — mirror-extension words, built so that palindromic prefixes recur along a geometric ladder of lengths — so Lemma 3.10 — which converts a single antidiagonal of width  $\geq 2$  into a width bound for all large  $s$  — has nothing to work with at  $k = 2$ . The existence half (Theorem 2.1), by contrast, is proved here for all  $k \geq 2$ .

The non-existence half rests on a finite additive-combinatorics statement which is of independent interest. A documented search of the subset-sum and additive-basis literature did not locate it, and it appears to be new; we set out the closest known results, and the caution proper to a negative novelty claim, in §1.4. For a finite multiset  $S$  of positive integers write  $\text{subset-sums}(S) := \{\sum_{x \in T} x : T \subseteq S\}$  (a selection of a sub-multiset), and for a finite set  $G$  of positive integers with  $M := \max G$  put

$$m(G) := \min \{ |S| : S \text{ a multiset drawn from } G, \\ \text{subset-sums}(S) \supseteq \text{an interval of } M \text{ consecutive integers} \}. \quad (1)$$

*Theorem B* (Bounded subset-sum interval theorem). For every finite  $G \subset \mathbb{Z}_{\geq 1}$  with  $|G| \geq 3$ ,  $\gcd(G) = 1$  and  $\max(G) = M$ , we have  $m(G) \leq M - 1$ .

We should say precisely what this bound does and does not assert. It is *not* the claim that  $M - 1$  is the exact value of  $m(G)$  for every  $G$ ; typically  $m(G)$  is far smaller. Rather,  $M - 1$  is (i) exactly the budget that the non-existence argument consumes at its binding corner  $k = d_2 = M$  (see the discussion after Lemma 3.14), and (ii) not improvable as a universal bound:

**Proposition 1.3** (Non-improvability). *Both hypotheses are used to their limit. The constant  $M - 1$  cannot be replaced by  $M - 2$  (indeed  $m(\{3, 4, 5\}) = 4 = M - 1$ ), and the cardinality hypothesis  $|G| \geq 3$  cannot be dropped: for  $G = \{2, 3\}$  (so  $\gcd = 1$ ,  $M = 3$ ) no multiset of size  $\leq M - 1 = 2$  has 3 consecutive integers among its subset sums.*

*Proof.* The multiset  $\{3, 3, 4, 5\}$ , of size 4, has subset sums  $\{0, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15\} \supseteq [3, 9]$ , a run of  $7 \geq 5$  consecutive integers, so  $m \leq 4$ . For the lower bound one checks the finitely many multisets drawn from  $\{3, 4, 5\}$  of size at most 3. There are 19 nonempty such multisets (3 of size 1, 6 of size 2, 10 of size 3), and none has 5 consecutive integers among its subset sums; the empty multiset, with subset sums  $\{0\}$ , is trivially excluded. In fact the longest subset-sum run achievable at size at most 3 is only 3, attained by  $\{3, 4, 5\}$  itself, whose subset sums are  $\{0, 3, 4, 5, 7, 8, 9, 12\}$  (longest run  $[3, 5]$ ). Hence  $m(\{3, 4, 5\}) = 4 = M - 1$ . For the cardinality claim, the five multisets drawn from  $\{2, 3\}$  of size  $\leq 2$  have subset-sum sets  $\{0, 2\}$ ,  $\{0, 3\}$ ,  $\{0, 2, 4\}$ ,  $\{0, 2, 3, 5\}$ ,  $\{0, 3, 6\}$ , none with a run of 3.  $\square$

Computation moreover suggests that the bound is attained along an infinite family: for the consecutive triples  $G = \{a, a + 1, a + 2\}$  we find  $m(G) = M - 1 = a + 1$  for every  $a$  with  $2 \leq a \leq 40$ . We have not proved this in general, and do not need it.

**1.4. Relation to the literature, and a caveat on novelty.** There is a substantial body of work on interval structure in *higher* sumsets. Nathanson [Na72] showed that for finite  $A$  with  $\min A = 0$  and  $\gcd(A) = 1$ , the  $m$ -fold sumset  $mA$  eventually consists of a head, a long block of consecutive integers, and a shifted tail; Granville and Shakan [GS20] described  $NA$  for large  $N$  and conjectured a sharp threshold, which Granville and Walker [GW21] proved; and Lev [Lev22] synthesizes this line, giving sharp estimates for the length of the central block. Lev's earlier theorem on consecutive integers in high-multiplicity sumsets [Lev10] is in the same vein.

These results are adjacent to Theorem B but do not appear to imply it, for a structural reason worth stating plainly. They concern  $mA$ , the  $m$ -fold sumset *with repetition*: nonnegative coefficients  $c_a$  with  $\sum_a c_a = m$  exactly, each  $c_a$  unbounded, and the interesting regime is  $m$  large. Theorem B is the opposite regime on both axes. It asks for a multiset  $S$  drawn from  $G$  of *bounded size*  $|S| \leq M - 1$ , and for its *subset sums* — 0/1 coefficients on the chosen multiset, each selected element used at most once — to contain  $M$  consecutive integers, uniformly in  $G$ . Neither the size bound nor the subset-sum (rather than  $m$ -fold) structure has a counterpart in the results above, and a statement about  $mA$  for large  $m$  says nothing about which sums are reachable using a *bounded* number of summands.

The closer literature is not the higher-sumset one but the work on *subset sums* proper. Here the object is right — 0/1 coefficients on a set or multiset — but the known theorems differ from Theorem B in a way that turns out to be decisive. The nearest results are the finite addition theorems for subset sums of Sárközy [Sa89, Sa94] and Lev [Lev03]: for a set  $A \subseteq [1, m]$  of size  $|A| \gtrsim \sqrt{m} \log m$ , the subset sums of  $A$  contain an arithmetic progression, with a bounded but generally  $> 1$  common difference (Lev [Lev03] sharpens the constant; Chen, Mao and Zhang [CMZ25] make the Sárközy results constructive, with a near-linear algorithm returning the progression and a representation of each of its terms, but still an arithmetic progression whose common difference need not be 1). Szemerédi and Vu [SV06], and the homogeneous strengthening of Conlon, Fox and Pham [CFP21], likewise obtain a long progression in the subset sums of a set  $A \subseteq [n]$  with  $|A| \geq c\sqrt{n}$ . All of these produce a *progression*, whose common difference cannot in general be forced to 1 (so not an interval of consecutive integers), from a *set* of *large* cardinality — the opposite of Theorem B, which asks for consecutive integers, from a generating set as small as  $|G| = 3$ , using a bounded multiset. (Lev’s earlier interval theorem [Lev97, Thm. 1] does give an interval of length  $\max(A) + 1$  — the “length equals the maximum” feature Theorem B shares — but inside the  $m$ -fold sumset  $mA$  with  $m$  large, not among the subset sums of a bounded multiset.) In short, none of these gives a *uniform* bound on the  $\ell^1$  size  $\sum_g n_g \leq M - 1$  of a bounded stock  $(n_g)_{g \in G}$  whose subset sums cover  $M$  consecutive integers, valid for every  $G$  down to  $|G| = 3$  — exactly the feature that the density-threshold and large-multiplicity theorems do not provide. Lev’s structural theorem on multisets with few subset sums [Lev99] is on the right object — a multiset, classified by its subset-sum set — but characterizes the multisets whose subset sums are *sparse*, an inverse problem to ours. Alon’s “Subset Sums” [Al87] treats the inverse (avoidance) extremal problem and is orthogonal.

On the strength of a documented search across these areas — subset sums of sets and multisets, completeness and complete sequences, the Frobenius and postage-stamp problems, interval structure in higher sumsets, and the bounded-coefficient literature — we have not found Theorem B, or a result that immediately implies it (even with a worse constant), in the literature, and it appears to be new: a bounded-stock (bounded  $\ell^1$ -size), uniform sharpening of the subset-sum interval phenomenon. We state this with appropriate caution. Novelty in a mature area is hard to establish negatively, our search was not exhaustive, and a specialist in additive combinatorics should confirm it before it is relied upon; the sharpest place to look is the immediate neighbourhood of Lev’s finite addition theorems, with which Theorem B shares the “interval of length  $\max$ ” feature. What we claim without qualification is that Theorem B is true, that we prove it, and that it is exactly what the non-existence half of Theorem A requires.

**1.5. Method.** The two halves of Theorem A are proved by unrelated methods.

For *existence* (§2) we place  $A$  inside a Beatty family  $A_{\gamma,c} = \{c + \lfloor i\gamma \rfloor : i \geq 1\}$  with  $\gamma$  slightly below  $d_2$ . Then the gaps of  $A$  automatically lie in  $\{d_2 - 1, d_2\}$ , and accumulated floor error confines  $kA$  to a union of half-open windows of width  $k$  hanging below the multiples of  $\gamma$ . Avoiding a given  $b$  becomes a statement about  $\gamma$  avoiding a union of “unsafe” intervals, and the crux (Lemma 2.2) is that one can *name a single gap* between consecutive unsafe intervals rather than enumerate them. A nested-interval argument then defeats all of  $B$  at once, and the lacunarity ratio  $192d_2$  is exactly what keeps the nested intervals from collapsing.

For *non-existence* (§3) the target is the property that  $kA$  is *tail-covering*: it eventually contains a full congruence class. A diagonalization (Lemma 3.1) shows that a single  $B$  then defeats every  $A$  simultaneously, so everything reduces to proving that every  $(d_1, d_2)$ -sequence with  $d_2 \leq k$  is tail-covering. This is done by cases on the *tail alphabet*  $G_\infty$  of gap values occurring infinitely often. One and two letter alphabets require a word-combinatorial analysis culminating in the Sturmian case (Lemma 3.11); alphabets with three or more letters are handled by an amortized

slot argument (Lemma 3.14) that reduces matters exactly to Theorem B, proved in §4 by a six-branch decision tree.

**1.6. Formal verification and the trust base.** The complete dichotomy — all three theorems, including the strong non-existence form — has been formalized in Lean 4 and machine-checked. The development is `sorry`-free and depends only on Lean’s three standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`); in particular it uses no `native_decide`, so every finite check is discharged by the trusted kernel rather than by compiled code. Section 6 states the formal encoding, argues its faithfulness to the informal problem, and records the trust base; Appendix C maps each load-bearing result of this paper to its Lean declaration.

We are explicit about what rests on what, and we keep three layers separate.

- The **prose** is the proof. A reader is asked to accept Theorem A on the strength of the argument in §§2–5, together with the certificate tables (defined and exemplified in Appendix B; listed in full in the supplement and the machine-readable exports). All infinite mathematical content — the whole of §2 and §3, and every “all  $M$ ” construction in §4 — is proved here in the text, and nothing in the argument is delegated to the formalization.
- The **finite layer** is exactly Tables A and B of Appendix B and the bounded scans of Appendix A, isolated as a single statement (Proposition A.1). Each table row is a self-certifying subset-sum witness, checkable by hand. The layer is corroborated by two independent Python harnesses that share no code, and re-decided inside the Lean kernel.
- The **Lean development** certifies the entire argument, finite and infinite together.

Accordingly, every step of this paper falls into exactly one of three auditable categories: an ordinary handwritten proof in the text; a finite certificate check with all data supplied (Appendices A–B); or a Lean-verified statement with an exact theorem name and source file (Appendix C). The Lean development is *not* used as an oracle for unstated mathematics.

A note on the origin of this work, and on the division of labour between the author and the automated systems used, is given in §7.

**1.7. Map of the proof.** The two halves are unrelated, and the reader may take them in either order.

|                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Theorem A</b> ( $r_k(d_1, d_2)$ exists $\iff d_2 \geq k + 1$ ) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| $d_2 \geq k + 1$ (existence, §2)                                  | Beatty family $A_{\gamma, c}$ (§2)<br>→ free-gap lemma (Lemma 2.2)<br>→ nested intervals → ratio $192d_2$ (Theorem 2.1)                                                                                                                                                                                                                                                                                                                                                                                                                             |
| $d_2 \leq k$ (non-existence, §3)                                  | certificate lemma (Lemma 3.1): one $B$ defeats all $A$ ,<br><i>provided</i> every admissible $A$ is <i>tail-covering</i> . Then split<br>on the tail alphabet $G_\infty$ :<br>$ G_\infty  = 1$ : arithmetic progression (Lemma 3.3)<br>$ G_\infty  = 2$ : eventually periodic (Lemma 3.5), or<br>width $\geq 2 \rightarrow$ sweep (Lemmas 3.8, 3.7), or<br>balanced $\rightarrow$ Prop. 3.9 $\rightarrow$ Sturmian (Lemma 3.11)<br>$ G_\infty  \geq 3$ : slot lemma (Lemma 3.14), which consumes<br><b>Theorem B</b> (§4, six branches D/P/L/E/T/B) |

The single point of contact between the two halves is Theorem B: it is what the slot lemma needs at its binding corner  $k = d_2 = M$ , and it is where the finite certificate data of Appendix B enters.

**1.8. Notation.** For a  $(d_1, d_2)$ -sequence  $A = \{a_1 < a_2 < \dots\}$  we write  $g_n := a_{n+1} - a_n \in [d_1, d_2] \cap \mathbb{Z}$  for its *gaps*, and  $p_0 := 0, p_n := g_1 + \dots + g_n$  for the *prefix sums*, so that  $a_{n+1} = a_1 + p_n$  and

$$kA = ka_1 + \{p_{i_1} + \dots + p_{i_k} : i_1, \dots, i_k \geq 0\}. \quad (2)$$

The *gap word* of  $A$  is the sequence  $(g_n)_{n \geq 1}$ , and  $G_\infty$  — the *tail alphabet* — denotes the set of gap values occurring infinitely often in it. We call  $A$  *tail-covering* if

$$kA \supseteq \{x \geq X_0 : x \equiv \rho \pmod{m}\} \quad \text{for some } m \geq 1, \rho, X_0.$$

Intervals of integers are written  $[u, v] = \{u, u+1, \dots, v\}$ ; for a real interval we say so explicitly. The quantity  $m(G)$  is as in (1).

## 2. EXISTENCE FOR $d_2 \geq k+1$

**Theorem 2.1.** *Let  $k \geq 2$ ,  $d_2 \geq k+1$  and  $1 \leq d_1 < d_2$ . Then  $r_k(d_1, d_2) \leq 192 d_2$ .*

The proof occupies the rest of this section. Fix  $B$  with  $b_{i+1} \geq r b_i$ , where  $r := 192 d_2$ . We construct  $A$  inside a Beatty family.

**2.1. Beatty cluster containment.** For real  $\gamma \in (d_2 - 1, d_2)$  and an integer  $c \geq 0$  set

$$A_{\gamma, c} := \{c + \lfloor i\gamma \rfloor : i \geq 1\}.$$

Its consecutive gaps are  $\lfloor (i+1)\gamma \rfloor - \lfloor i\gamma \rfloor \in \{d_2 - 1, d_2\} \subseteq [d_1, d_2]$ , so  $A_{\gamma, c}$  is a  $(d_1, d_2)$ -sequence for every such  $\gamma$  and  $c$ . For any  $i_1, \dots, i_k \geq 1$  with  $s := \sum_t i_t$  we have, from  $i_t \gamma - 1 < \lfloor i_t \gamma \rfloor \leq i_t \gamma$ ,

$$\sum_t \lfloor i_t \gamma \rfloor \in (s\gamma - k, s\gamma],$$

and therefore

$$kA_{\gamma, c} \subseteq \bigcup_{s \geq k} (kc + s\gamma - k, kc + s\gamma]. \quad (3)$$

Consequently, if for a given  $b$  no integer  $s \geq 1$  satisfies  $s\gamma \in [b - kc, b - kc + k]$ , then  $b \notin kA_{\gamma, c}$ . Two deliberate weakenings make this criterion easier to verify than (3) requires. First, we quantify over all  $s \geq 1$  rather than only the  $s \geq k$  that actually occur in (3); excluding more values of  $s$  is a stronger — hence safe — demand, and it frees the argument from tracking which  $s$  are realizable. Second, we use the *closed* interval  $[b - kc, b - kc + k]$  in place of the half-open window of (3), which again only strengthens the requirement. No irrationality of  $\gamma$  is needed anywhere.

**2.2. The free-gap lemma.** This is the engine of the construction.

**Lemma 2.2** (Free gap). *Let  $t \geq 32d_2^2$  and let  $I \subseteq [d_2 - \frac{1}{2}, d_2]$  be a closed real interval with  $|I| \geq 4d_2^2/t$ . Then  $I$  contains a closed subinterval  $I'$  with  $|I'| \geq d_2/(48t)$  such that every  $\gamma \in I'$  satisfies  $s\gamma \notin [t, t+k]$  for all integers  $s \geq 1$ .*

*Proof.* Write  $I = [lo, hi]$ , so  $d_2 - \frac{1}{2} \leq lo \leq hi \leq d_2$  and  $hi - lo \geq 4d_2^2/t$ . A real  $\gamma$  satisfies  $s\gamma \notin [t, t+k]$  for all  $s \geq 1$  exactly when  $\gamma$  avoids the *unsafe set*

$$U := \bigcup_{s \geq 1} [t/s, (t+k)/s].$$

Rather than count the unsafe intervals meeting  $I$ , we name a single gap between two of them. Put

$$s_1 := \lceil t/hi \rceil \quad (\text{the least } s \text{ with } t/s \leq hi), \quad R := t/s_1, \quad L := (t+k)/(s_1+1),$$



and let  $J := (L, R)$ , the gap between the  $s_1$ -th and  $(s_1+1)$ -st unsafe intervals. From  $t/hi \leq s_1 < t/hi + 1$ , together with  $hi \in [d_2 - \frac{1}{2}, d_2]$  and  $t \geq 32d_2^2$ , one gets

$$\frac{t}{d_2} \leq s_1, \quad s_1 \geq 32d_2, \quad s_1 + 1 \leq \frac{5t}{d_2}. \quad (4)$$

(i)  $J$  avoids  $U$ . If  $s \leq s_1$  then  $t/s \geq t/s_1 = R$ , so any  $\gamma < R$  has  $s\gamma < t$ . If  $s \geq s_1 + 1$  then  $(t+k)/s \leq (t+k)/(s_1+1) = L$ , so any  $\gamma > L$  has  $s\gamma > t+k$ . Hence a single  $\gamma \in J$  misses *every* interval  $[t/s, (t+k)/s]$  at once; no enumeration of a relevant range of  $s$  is required.

(ii)  $|J| \geq d_2/(16t)$ . We have  $R-L = (t-ks_1)/(s_1(s_1+1))$ . Since  $s_1 \leq t/hi+1 \leq t/(d_2-\frac{1}{2})+1$  and  $k \leq d_2-1$ ,

$$\begin{aligned} \left(k + \frac{15}{32}\right)s_1 &\leq \left(d_2 - \frac{17}{32}\right)\left(\frac{t}{d_2 - \frac{1}{2}} + 1\right) \\ &= t - \frac{t}{32(d_2 - \frac{1}{2})} + \left(d_2 - \frac{17}{32}\right) \leq t, \end{aligned}$$

the last step because  $t \geq 32d_2^2$  gives  $t/(32(d_2 - \frac{1}{2})) = d_2^2/(d_2 - \frac{1}{2}) \geq d_2 + \frac{1}{2} > d_2 - \frac{17}{32}$ . Thus  $t - ks_1 \geq \frac{15}{32}s_1$ , and with (4),

$$R - L \geq \frac{15}{32} \cdot \frac{1}{s_1 + 1} \geq \frac{15}{32} \cdot \frac{d_2}{5t} = \frac{3}{32} \cdot \frac{d_2}{t} \geq \frac{d_2}{16t}.$$

(iii)  $J \subseteq I$ . First  $R = t/s_1 \leq hi$  by the choice of  $s_1$ , and  $s_1 < t/hi+1$  gives  $R > t/(t/hi+1) > hi - hi^2/t \geq hi - d_2^2/t$ . Also  $R - L \leq t/(s_1(s_1+1)) \leq t/(t/hi)^2 \leq d_2^2/t$ , so

$$L = R - (R - L) \geq (hi - d_2^2/t) - d_2^2/t = hi - 2d_2^2/t \geq hi - \frac{1}{2}(hi - lo) \geq lo.$$

Hence  $[L, R] \subseteq [lo, hi] = I$ .

Now take  $I'$  to be the closed middle third of  $J$ . Then  $I' \subseteq I$ , every  $\gamma \in I'$  avoids  $U$  by (i), and  $|I'| = |J|/3 \geq d_2/(48t)$  by (ii).  $\square$

**2.3. Small elements of  $B$ , and the nested intervals.** Since  $B$  is infinite and increasing, we may fix an index  $j_0$  with  $b_{j_0} \geq 4d_2^2$ . Set  $c := b_{j_0} + 1$ . Then  $\min kA_{\gamma,c} \geq k(c + d_2 - 1) > kc > b_{j_0}$ , so every  $b_i \leq b_{j_0}$  lies below  $\min kA$  and is avoided automatically. For  $i > j_0$  put  $t_i := b_i - kc$ ; these are positive, since  $b_i \geq b_{j_0+1} \geq 192d_2b_{j_0}$  while  $kc = k(b_{j_0} + 1) \leq d_2(b_{j_0} + 1)$ , and  $192d_2b_{j_0} > d_2(b_{j_0} + 1)$  for  $b_{j_0} \geq 1$ . Since  $(x - \tau)/(y - \tau) \geq x/y$  whenever  $x \geq y > \tau \geq 0$ , the shifted targets retain the ratio:

$$\frac{t_{i+1}}{t_i} \geq \frac{b_{i+1}}{b_i} \geq 192d_2,$$

and moreover  $t_{j_0+1} = b_{j_0+1} - kc \geq 192d_2b_{j_0} - d_2(b_{j_0} + 1) \geq 96d_2b_{j_0} \geq 384d_2^3 \geq 32d_2^2$ .

*Proof of Theorem 2.1.* Start with the closed real interval  $I_0 := [d_2 - \frac{1}{2}, d_2 - \frac{1}{4}]$ , which has  $|I_0| = \frac{1}{4} \geq 4d_2^2/t_{j_0+1}$  by the previous paragraph. Applying Lemma 2.2 with  $t = t_{j_0+1}$  produces a closed  $I_1 \subseteq I_0$ , safe for  $t_{j_0+1}$ , with  $|I_1| \geq d_2/(48t_{j_0+1})$ . Now induct for  $j \geq 1$ : suppose  $I_j$  is safe for the targets  $t_{j_0+1}, \dots, t_{j_0+j}$  and satisfies  $|I_j| \geq d_2/(48t_{j_0+j})$  (both well defined, as  $j \geq 1$ ). Apply Lemma 2.2 with  $t = t_{j_0+j+1}$ : its hypothesis  $|I_j| \geq 4d_2^2/t$  reduces to

$$t_{j_0+j+1} \geq 192d_2t_{j_0+j},$$

which is exactly what the ratio  $r = 192d_2$  supplies. This produces a closed  $I_{j+1} \subseteq I_j$ , safe also for  $t_{j_0+j+1}$ , with  $|I_{j+1}| \geq d_2/(48t_{j_0+j+1})$ .

The  $I_j$  are nested nonempty closed subintervals of  $I_0$ , so by the nested interval theorem (equivalently, by compactness of  $I_0$ ) the intersection  $\bigcap_j I_j$  is nonempty. Fix  $\gamma^* \in \bigcap_j I_j$ ; it is safe for every  $t_i$  simultaneously. By (3) and the criterion following it,  $(kA_{\gamma^*,c}) \cap B = \emptyset$ . As  $A_{\gamma^*,c}$  is a  $(d_1, d_2)$ -sequence, this proves  $r_k(d_1, d_2) \leq 192d_2$ .  $\square$

*Remark 2.3.* Theorem 2.1 is stated and proved here for all  $k \geq 2$ . The Lean formalization covers  $k \geq 3$  — the range of the problem and of the dichotomy Theorem A — so the case  $k = 2$  is established in prose only. This is deliberate: the problem fixes  $k \geq 3$ , and the formal development carries exactly the problem’s hypotheses, so that every headline Lean theorem (`erdos_1112`, `erdos_1112_existence_bound`, `erdos_1112_strong_nonexistence`) reads with  $3 \leq k$ , matching the statement on `erdosproblems.com`.

**Glossary of terms coined in §§3–4.** Several constructions below are given short names. They are collected here with their formal content.

| Term                                    | Formal content                                                                       |
|-----------------------------------------|--------------------------------------------------------------------------------------|
| <i>tail-covering</i>                    | $kA \supseteq \{x \geq X_0 : x \equiv \rho \pmod{m}\}$ for some $m, \rho, X_0$       |
| <i>tail alphabet</i> $G_\infty$         | the gap values occurring infinitely often in $(g_n)$                                 |
| <i>gap word</i>                         | the sequence $(g_n)$ of consecutive differences of $A$                               |
| <i>antidiagonal width</i> $W_2(\sigma)$ | $\max\{q_i + q_j : i + j = \sigma\} - \min\{q_i + q_j : i + j = \sigma\}$            |
| <i>fine slot</i>                        | a summand index free to take one of two values, $n_t - 1$ or $n_t$                   |
| <i>coarse dial</i>                      | the last summand $p_n$ , whose steps are $\leq M$                                    |
| <i>hard core</i>                        | triples $\{a, b, M\}$ , $\gcd(a, b) = 1$ , $\delta = a + b - M \geq 2$ (§4.3)        |
| <i>frame</i>                            | a box $(Y, Z) \bmod a$ covering $\mathbb{Z}_a$ , lifted by copies of $a$ (Lemma 4.3) |
| <i>budget</i>                           | the size $x + y + z$ of a multiset, to be compared with $M - 1$                      |
| <i>staircase</i>                        | the level-by-level decomposition of subset sums (Lemma 4.4)                          |
| <i><math>\lambda</math>-lift</i>        | the descent $b \mapsto b + a$ , $M \mapsto M + a$ within a class (Lemma 4.6)         |

### 3. NON-EXISTENCE FOR $d_2 \leq k$ : REDUCTION TO THEOREM B

Throughout this section  $k \geq 3$  and  $d_2 \leq k$ .

#### 3.1. The certificate.

**Lemma 3.1** (Certificate). *Fix any enumeration  $(m_i, \rho_i)_{i \geq 1}$  of all pairs  $\{(m, \rho) : m \geq 1, 0 \leq \rho < m\}$  in which each pair occurs infinitely often. Let  $R : \mathbb{N} \rightarrow \mathbb{N}$  be an arbitrary prescribed ratio sequence (no monotonicity is assumed). Set  $b_1 := 1$ , and for  $i \geq 1$  let*

$$b_{i+1} := \text{the least integer} > \max(R(i)b_i, b_i, i) \text{ that is } \equiv \rho_{i+1} \pmod{m_{i+1}}.$$

*Then  $B = \{b_1 < b_2 < \dots\}$  satisfies  $b_{i+1} \geq R(i)b_i$  for every  $i$ . If moreover every  $(d_1, d_2)$ -sequence  $A$  is tail-covering, then this single  $B$  defeats them all:  $(kA) \cap B \neq \emptyset$  for every  $(d_1, d_2)$ -sequence  $A$ .*

*Proof.* Each  $b_{i+1}$  exists (an arithmetic progression modulo  $m_{i+1}$  meets every half-line), and by construction  $b_{i+1} > b_i$  and  $b_{i+1} > R(i)b_i$ , so  $B$  is strictly increasing, positive, and lacunary with the prescribed ratios. Note that  $R$  is arbitrary: no monotonicity is used, so a prescribed sequence need not be replaced by its nondecreasing envelope.

Let  $A$  be a  $(d_1, d_2)$ -sequence and pick  $(m, \rho, X_0)$  with  $kA \supseteq \{x \geq X_0 : x \equiv \rho \pmod{m}\}$ . Choose an index  $i \geq 2$  with  $(m_i, \rho_i) = (m, \rho)$  and  $b_i \geq X_0$ ; this is possible because the pair  $(m, \rho)$  recurs infinitely often and  $b_i \geq i \rightarrow \infty$  ( $B$  is strictly increasing of positive integers). Then  $b_i \equiv \rho \pmod{m}$  and  $b_i \geq X_0$ , so  $b_i \in kA$ .  $\square$

So it suffices to prove:

*every  $(d_1, d_2)$ -sequence with  $d_2 \leq k$  is tail-covering.*

Full congruence-class tails are genuinely needed here, rather than merely “all large integers”: the modulus can exceed  $d_2$ . For instance, gaps  $3, 5, 3, 5, \dots$  with  $k = 5$  give exactly the six classes  $5a_1 + \{0, 1, 3, 4, 6, 7\} \pmod{8}$ .



### 3.2. Elementary reductions.

**Lemma 3.2** (Tail). *Let  $T$  be an index past which all gaps of  $A$  lie in  $G_\infty$ . Since  $kA$  contains the  $k$ -fold sums of  $A \cap [a_T, \infty)$ , it suffices to prove the tail-covering property for the tail. We do so silently from now on.*

**Lemma 3.3** (One letter). *If  $G_\infty = \{\delta\}$  then the tail of  $A$  is an arithmetic progression and  $kA \supseteq \{ka_T + \delta j : j \geq 0\}$ , so  $A$  is tail-covering with  $m = \delta$ .*  $\square$

**Lemma 3.4** (Rescaling). *If  $g := \gcd(G_\infty) > 1$ , write the tail of  $A$  as  $a_T + gA'$ , where  $A'$  has gap alphabet  $G_\infty/g$ , of  $\gcd 1$  and maximum  $\leq d_2/g$ . If the  $k$ -fold sums of  $A'$  are tail-covering modulo  $m'$ , then  $kA$  is tail-covering modulo  $gm'$ . Hence we may and do assume  $\gcd(G_\infty) = 1$ .*  $\square$

**Lemma 3.5** (Eventually periodic). *If the gap word is eventually periodic then the tail of  $A$  is a full pattern  $(C + P\mathbb{Z}) \cap [T', \infty)$  for a finite set  $C$  and period-sum  $P$ . For each  $k$ -element multiset of  $C$  with sum  $\sigma$ , the sumset  $kA$  then contains the entire class tail  $\sigma + P\mathbb{Z}$  beyond a point; so  $A$  is tail-covering.*  $\square$

**3.3. The two-letter core.** Assume now  $G_\infty = \{\delta < \delta'\}$  with  $\gcd(\delta, \delta') = 1$  (Lemma 3.4), both letters occurring infinitely often, and set  $e := \delta' - \delta \geq 1$ . Since every gap lies in  $[d_1, d_2]$  and  $d_2 \leq k$ , the larger letter satisfies

$$\delta' = \delta + e \leq d_2 \leq k. \quad (5)$$

(The tail alphabet need not attain  $d_2$ ; only the inequality (5) is used.) Write the gaps as  $g_n = \delta + eh_n$  with  $h_n \in \{0, 1\}$ , and let  $q_n := h_1 + \cdots + h_n$  be the counting function of the binary gap word  $h$ . By (2), every  $x \in kA - ka_1$  has the form

$$x = \delta s + ew, \quad s = \sum_t i_t, \quad w = \sum_t q_{i_t},$$

so that  $w$  ranges over  $W_s := \{\sum_t q_{i_t} : \sum_t i_t = s\}$ .

**Lemma 3.6** (Interval property). *For each  $s$ , the set  $W_s$  is a full integer interval  $[w^-(s), w^+(s)]$ , and  $w^\pm(s+1) - w^\pm(s) \in \{0, 1\}$ .*

*Proof.* The set of  $k$ -multisets of nonnegative integers with fixed sum  $s$  is connected under the moves  $(i_a, i_b) \mapsto (i_a - 1, i_b + 1)$ , and each such move changes  $\sum_t q_{i_t}$  by  $h_{i_b+1} - h_{i_a} \in \{-1, 0, 1\}$ . A path from a minimizer to a maximizer therefore visits a set of integer values with no gap of size  $\geq 2$ , so  $W_s$  is an interval. The increment claim follows by comparing multisets for  $s$  and  $s+1$  differing in a single index.  $\square$

**Lemma 3.7** (Sweep). *If  $w^+(s) - w^-(s) \geq d_2 - 1$  for all  $s \geq S_0$ , then  $kA - ka_1$  contains all sufficiently large integers; in particular  $A$  is tail-covering with  $m = 1$ .*

*Proof.* Let  $x$  be large. Representations  $x = \delta s + ew$  force  $s \equiv x\delta^{-1} \pmod{e}$ , so the admissible  $s$  form an arithmetic progression of common difference  $e$ , and for each such  $s$  the value  $w(s) := (x - \delta s)/e$  increases by exactly  $\delta$  when  $s$  decreases by  $e$ . The windows  $[w^-(s), w^+(s)]$  are nondecreasing in  $s$  with increments in  $[0, e]$  per admissible step (Lemma 3.6). For the largest admissible  $s$  with  $w(s) \geq 0$  we have  $w(s) < \delta$ ; and  $\delta \leq w^+(s)$  for all large  $s$ , since the letter 1 occurs infinitely often and hence  $w^+(s) \geq q_s \rightarrow \infty$  — enlarging  $S_0$  if necessary, we assume  $w^+(s) \geq \delta$  for all  $s \geq S_0$ . At the other end, for the smallest admissible  $s \geq S_0$  we have  $w(s) = (x - \delta s)/e \geq (x - \delta(S_0 + e))/e$ , which exceeds  $w^+(s) \leq s$  once  $x$  is large (explicitly, once  $x > \delta(S_0 + e) + e(S_0 + e)$ ); so  $w(s) > w^+(s)$  there.

Suppose no admissible  $s \geq S_0$  had  $w(s) \in [w^-(s), w^+(s)]$ . Since  $w(\cdot)$  decreases along the admissible progression while the windows increase, there is then a crossing pair  $s', s' + e$  with

$w(s' + e) < w^-(s' + e)$  and  $w(s') > w^+(s')$ . We bound  $s'$  from below explicitly. The counting function satisfies  $q_i \leq i$ , so for any  $k$  indices with  $\sum_t i_t = s$  we have  $\sum_t q_{i_t} \leq s$ ; hence  $w^+(s) \leq s$  for every  $s$ . At the index  $s' + e$  the inequality has reversed, so

$$\frac{x - \delta(s' + e)}{e} = w(s' + e) \leq w^-(s' + e) \leq w^+(s' + e) \leq s' + e,$$

which rearranges to  $s' + e \geq x/(\delta + e)$ , that is

$$s' \geq \frac{x}{\delta + e} - e.$$

In particular  $s' \geq S_0$  as soon as  $x \geq (\delta + e)(S_0 + e)$ , which we assume. But then

$$w^+(s') < w(s') = w(s' + e) + \delta \leq w^-(s' + e) - 1 + \delta \leq w^-(s') + e - 1 + \delta,$$

that is,  $w^+(s') - w^-(s') \leq (\delta + e) - 2 = \delta' - 2 \leq d_2 - 2$ , contradicting the hypothesis (which is where width  $\geq d_2 - 1 \geq \delta' - 1$  is used). So some admissible  $s \geq S_0$  has  $w(s) \in [w^-(s), w^+(s)]$ , and by Lemma 3.6 that value is attained.  $\square$

For  $\sigma \geq 0$  define the *two-index antidiagonal width*

$$W_2(\sigma) := \max\{q_i + q_j : i + j = \sigma\} - \min\{q_i + q_j : i + j = \sigma\}.$$

**Lemma 3.8** (Width dichotomy).

(a) Suppose  $W_2(\sigma_0) \geq 2$  for some  $\sigma_0$ . Disjoint index-pairs placed on antidiagonals contribute widths additively, and:

- $k$  odd:  $(k-1)/2$  pairs at  $\sigma_0$  plus one leftover index give  $w^+(s) - w^-(s) \geq k-1 \geq d_2-1$  for all  $s \geq \frac{k-1}{2}\sigma_0$ , using exactly  $k$  indices.
- $k$  even,  $d_2 \leq k-1$ :  $(k/2-1)$  pairs at  $\sigma_0$  plus one free pair at  $\tau := s - (k/2-1)\sigma_0$  give width  $\geq (k-2) + W_2(\tau) \geq k-2 \geq d_2-1$ .
- $k$  even,  $d_2 = k$  (boundary): three possibilities, treated in turn. If the gap word is eventually periodic, Lemma 3.5 applies. Otherwise, if some tail of the gap word is balanced, pass to that tail (Lemma 3.2) and apply the Morse–Hedlund dichotomy as in part (b): the tail is ultimately periodic (Lemma 3.5) or Sturmian (Lemma 3.11), and either way  $kA$  is tail-covering with no width argument needed. Otherwise every tail of the gap word is unbalanced, and Lemma 3.10 supplies width  $\geq k-1$  for all large  $s$ .

In every branch ending with a width bound, Lemma 3.7 then gives tail-covering.

(b) Otherwise  $W_2(\sigma) \leq 1$  for all  $\sigma$ . Then for any equal-length windows  $(i, i']$  and  $(j', j]$  one has  $i + j = i' + j'$ , whence  $|(q_{i'} - q_i) - (q_j - q_{j'})| = |q_{i'} + q_{j'} - q_i - q_j| \leq 1$ : the word  $h$  is balanced. By Proposition 3.9 a balanced one-sided infinite word is either ultimately periodic (Lemma 3.5) or, after passing to a tail, mechanical with irrational slope, and the latter is Lemma 3.11.

The parity split in (a) is essential: when  $k$  is even,  $\lfloor k/2 \rfloor$  pairs plus a leftover index would consume  $k+1$  indices.

This dichotomy is the classical Morse–Hedlund theorem [MH40] (see also [Lo02, Ch. 2]), and a reader content to cite it may pass directly to Lemma 3.11. We nonetheless give a proof, for two reasons: we need only one direction of it, and the argument below is the one our formalization carries out — it is elementary, and its only analytic input is Dirichlet’s approximation theorem. No density, equidistribution or three-distance statement is used anywhere.

**Proposition 3.9** (Balanced classification). *Let  $h : \mathbb{N} \rightarrow \{0, 1\}$  be balanced, with  $q_n := h_1 + \dots + h_n$  and  $q_0 := 0$ . Then  $h$  is either eventually periodic, or — after passing to a tail — satisfies  $q_n = \lfloor n\alpha + \beta \rfloor$  for every  $n \geq 1$ , for some irrational  $\alpha \in (0, 1)$  and some  $\beta \in \mathbb{R}$ .*

*Proof. Step 1: near-additivity.* For all  $n, m \geq 0$ ,

$$|q_{n+m} - q_n - q_m| \leq 1. \quad (6)$$

Indeed  $q_{n+m} - q_n$  counts the letters 1 in the window  $h_{n+1} \cdots h_{n+m}$ , of length  $m$ , while  $q_m$  counts them in  $h_1 \cdots h_m$ , also of length  $m$ ; balancedness bounds the difference of these two counts by 1.

*Step 2: the slope exists, and  $|q_n - n\alpha| \leq 1$ .* By (6) the sequence  $a_n := q_n + 1$  is subadditive,  $a_{n+m} \leq a_n + a_m$ , and the sequence  $b_n := q_n - 1$  is superadditive,  $b_{n+m} \geq b_n + b_m$ . By Fekete's lemma both  $a_n/n$  and  $b_n/n$  converge, to  $\inf_{n \geq 1} a_n/n$  and  $\sup_{n \geq 1} b_n/n$  respectively. Since  $a_n/n$  and  $b_n/n$  differ by  $2/n \rightarrow 0$ , the two limits coincide; call the common value  $\alpha$ . Then  $q_n/n \rightarrow \alpha$ , and

$$\alpha = \sup_{n \geq 1} \frac{q_n - 1}{n} = \inf_{n \geq 1} \frac{q_n + 1}{n},$$

so that for every  $n \geq 1$  we have  $(q_n - 1)/n \leq \alpha \leq (q_n + 1)/n$ , i.e.

$$|D_n| \leq 1, \quad D_n := q_n - n\alpha \quad (n \geq 0), \quad (7)$$

with  $D_0 = 0$ . Also  $\alpha \in [0, 1]$ , since  $0 \leq q_n \leq n$ .

*Step 3: the oscillation bound.* For all  $n \geq 0$  and  $\Delta \geq 1$ ,

$$|D_{n+\Delta} - D_n| \leq 1. \quad (8)$$

Fix  $\Delta$  and write  $W(j) := q_{j+\Delta} - q_j$  for the number of 1s in the length- $\Delta$  window starting at position  $j + 1$ . Balancedness gives  $|W(j) - W(n)| \leq 1$  for all  $j$ . Summing over the  $t$  consecutive disjoint blocks starting at  $0, \Delta, \dots, (t-1)\Delta$  telescopes:

$$q_{t\Delta} = \sum_{i=0}^{t-1} W(i\Delta) \in [t(W(n) - 1), t(W(n) + 1)].$$

Dividing by  $t\Delta$  and letting  $t \rightarrow \infty$ , where  $q_{t\Delta}/(t\Delta) \rightarrow \alpha$  by Step 2, gives  $(W(n) - 1)/\Delta \leq \alpha \leq (W(n) + 1)/\Delta$ , i.e.  $|W(n) - \Delta\alpha| \leq 1$ . Since  $D_{n+\Delta} - D_n = W(n) - \Delta\alpha$ , this is (8). (Taking  $n = 0$  recovers (7).)

*Step 4:  $\alpha$  rational  $\Rightarrow h$  eventually periodic.* Let  $\alpha = p/q$  in lowest terms and put  $c_n := q q_n - p n = q D_n$ , an integer with  $|c_n| \leq q$  by (7). With  $w_n := q_{n+q} - q_n$  the length- $q$  window counts,

$$c_{n+q} - c_n = q(w_n - p).$$

Balancedness confines the  $w_n$  to at most two consecutive values, and  $p$  is among them: if  $w_n \geq p + 1$  for every  $n$  then  $c_{n+q} - c_n \geq q$  for every  $n$ , so  $c$  would increase without bound along each residue class modulo  $q$ , contradicting  $|c_n| \leq q$ ; symmetrically,  $w_n \leq p - 1$  for every  $n$  would drive  $c$  to  $-\infty$ . Hence either  $w_n \in \{p, p + 1\}$  for all  $n$ , or  $w_n \in \{p - 1, p\}$  for all  $n$ .

In the first case  $c_{n+q} - c_n \in \{0, q\}$ , so along each residue class modulo  $q$  the sequence  $c$  is nondecreasing; being confined to  $[-q, q]$ , it is eventually constant on that class. In the second case  $c$  is nonincreasing, with the same conclusion. Either way  $w_n = p$  for all large  $n$ , i.e.  $q_{n+q} = q_n + p$  for all large  $n$ , whence

$$h_{n+q+1} = q_{n+q+1} - q_{n+q} = (q_{n+1} + p) - (q_n + p) = h_{n+1}$$

for all large  $n$ : the word  $h$  is eventually periodic, of period  $q$ .

*Step 5:  $\alpha$  irrational  $\Rightarrow h$  is mechanical on a tail.* Since  $\alpha$  is irrational,  $\alpha \notin \{0, 1\}$ , so  $\alpha \in (0, 1)$ . Put

$$\beta := \sup_{n \geq 0} D_n,$$

which is finite by (7). By the oscillation bound (8),  $D_n \geq \beta - 1$  for every  $n$ : indeed for any  $n$  and any  $m$ ,  $D_m - D_n \leq 1$ , and taking the supremum over  $m$  gives  $\beta - D_n \leq 1$ .

Now observe that, for a given  $n$ ,

$$q_n = \lfloor n\alpha + \beta \rfloor \iff n\alpha + \beta - 1 < q_n \leq n\alpha + \beta \iff \beta - 1 < D_n \leq \beta.$$

The right-hand inequality  $D_n \leq \beta$  holds by the definition of  $\beta$ , and the left-hand one holds in the weak form  $D_n \geq \beta - 1$  by the previous paragraph. So the identity can fail only when  $D_n = \beta - 1$  exactly — equivalently, when  $n\alpha + \beta = q_n + 1$  is an integer, i.e.  $\{n\alpha + \beta\} = 0$ . If two indices  $n_1 \neq n_2$  both had  $n_i\alpha + \beta \in \mathbb{Z}$ , then  $(n_1 - n_2)\alpha \in \mathbb{Z}$ , which is impossible for irrational  $\alpha$ . So there is *at most one* exceptional index; call it  $n_0$  if it exists, and set  $n_0 := 0$  otherwise (recall  $q_0 = 0 = \lfloor \beta \rfloor$  can be arranged, and in any case the index 0 is discarded when we pass to the tail). Then

$$q_n = \lfloor n\alpha + \beta \rfloor \quad \text{for every } n \geq 1 \text{ with } n \neq n_0.$$

Finally we pass to the tail beyond  $n_0$  (Lemma 3.2). The tail word  $h'_j := h_{n_0+j}$  has counting function  $q'_j = q_{n_0+j} - q_{n_0}$ , and for every  $j \geq 1$  (so that  $n_0 + j \neq n_0$ ),

$$q'_j = \lfloor (n_0 + j)\alpha + \beta \rfloor - q_{n_0} = \lfloor j\alpha + (n_0\alpha + \beta - q_{n_0}) \rfloor = \lfloor j\alpha + \beta' \rfloor, \quad \beta' := \beta - D_{n_0},$$

using that  $q_{n_0}$  is an integer. So the tail is mechanical with slope  $\alpha$  and intercept  $\beta'$  for every  $j \geq 1$ . (Nothing is claimed at  $j = 0$ , where  $q'_0 = 0$  by definition while  $\lfloor \beta' \rfloor$  may be 1. Only indices  $j \geq 1$  are used in Lemma 3.11, whose summand indices  $i_t$  are all  $\geq 1$ .)  $\square$

Everything above rests on the near-additivity (6) and the discrepancy bound (7), not on any density or equidistribution statement. Similarly, the uniform syndeticity used in Step 1 of Lemma 3.11 below is obtained from a single Dirichlet approximation (a bounded circle walk), not from the three-distance theorem. This is the route carried out in the formalization (`disc_osc_le_one`, `mechanical_tail`, `walk_enters`).

**Lemma 3.10** (Boundary width;  $k$  even,  $k \geq 4$ ). *Let  $h$  use both letters infinitely often, let  $W_2(\sigma_0) \geq 2$  for some  $\sigma_0$ , let  $h$  be not eventually periodic, and suppose every tail of  $h$  is unbalanced. Then  $w^+(s) - w^-(s) \geq k - 1$  for all sufficiently large  $s$ .*

*Proof.* First we check that unbalancedness of *every* tail does give  $W_2(\sigma) \geq 2$  for infinitely many  $\sigma$ . Suppose not, so that  $W_2(\sigma) \leq 1$  for every  $\sigma \geq \sigma^*$ . Let  $(i, i']$  and  $(j', j]$  be any two equal-length windows lying beyond  $\sigma^*$ ; then  $i + j = i' + j'$ , so  $W_2(i + j) \leq 1$  gives  $|q_{i'} + q_{j'} - q_i - q_j| \leq 1$ , i.e.

$$|(q_{i'} - q_i) - (q_j - q_{j'})| \leq 1.$$

That is precisely the statement that the tail of  $h$  beyond  $\sigma^*$  is balanced, contradicting the hypothesis that every tail is unbalanced. Hence  $W_2(\sigma) \geq 2$  for infinitely many  $\sigma$ .

Fix two of them,  $\sigma_0 < \sigma_1$ , and put  $\Delta := \sigma_1 - \sigma_0$ . For large  $s$  consider two configuration families, each using exactly  $k$  indices:

- (1)  $(k/2 - 1)$  extremal pairs at  $\sigma_0$ , plus one free pair at  $\tau := s - (k/2 - 1)\sigma_0$ , giving width  $\geq (k - 2) + W_2(\tau)$ ;
- (2)  $(k/2 - 2)$  extremal pairs at  $\sigma_0$ , one extremal pair at  $\sigma_1$ , plus one free pair at  $\tau - \Delta$ , giving width  $\geq (k - 2) + W_2(\tau - \Delta)$ . (Here  $k/2 - 2 \geq 0$  uses  $k \geq 4$ .)

If  $w^+(s) - w^-(s) \leq k - 2$ , both force  $W_2(\tau) = W_2(\tau - \Delta) = 0$ . But  $W_2(\tau) = 0$  means  $q_i + q_{\tau-i}$  is constant in  $i$ , i.e.  $h_{i+1} = h_{\tau-i}$  for  $0 \leq i \leq \tau - 1$ : the prefix  $h[1..\tau]$  is a palindrome. Likewise  $h[1..\tau - \Delta]$  is a palindrome. A palindromic prefix of a palindrome is also its suffix, so  $h[1..\tau - \Delta]$  is a border of  $h[1..\tau]$ , and hence  $h[1..\tau]$  has period  $\Delta$ : this is the classical border–period duality (a word  $w$  with a border of length  $\ell$  has period  $|w| - \ell$ , since the border condition  $w_i = w_{i+|w|-\ell}$  holds for all  $i \leq \ell$ ; see [Lo02]). If the width bound failed for infinitely many  $s$  then —  $\Delta$  being fixed while  $\tau(s)$  strictly increases — the word  $h$  would have period  $\Delta$ , contradicting non-periodicity.  $\square$

**Lemma 3.11** (Sturmian case). *Let the two-letter tail of §3.3 have  $q_n = \lfloor n\alpha + \beta \rfloor$  for an irrational  $\alpha \in (0, 1)$  and some  $\beta \in \mathbb{R}$ . Then  $kA - ka_1$  contains all sufficiently large integers, so  $A$  is tail-covering with  $m = 1$ .*

*Proof.* We keep the notation of §3.3:  $e = \delta' - \delta \geq 1$ ,  $\gcd(\delta, e) = 1$ , and  $\delta' = \delta + e \leq k$  by (5). Every  $x \in kA - ka_1$  has the form  $x = \delta s + ew$  with  $s = \sum_t i_t$  (all  $i_t \geq 1$ , as we work in the tail) and  $w = \sum_t q_{i_t} \in W_s$ .

*Step 0: the window carrying  $W_s$ .* From  $q_n = n\alpha + \beta - \{n\alpha + \beta\}$ ,

$$\sum_t q_{i_t} = \left( \alpha \sum_t i_t + k\beta \right) - \sum_t \{i_t\alpha + \beta\} = X_s - \sum_t \{i_t\alpha + \beta\}, \quad X_s := s\alpha + k\beta.$$

Each  $\{i_t\alpha + \beta\} \in [0, 1)$ , so  $\sum_t \{i_t\alpha + \beta\} \in [0, k)$  and hence

$$W_s \subseteq (X_s - k, X_s] \cap \mathbb{Z}. \quad (9)$$

*Step 1: uniform syndeticity.* For every arc  $J = (u, v) \subseteq (0, 1)$  with  $v - u > 0$  there is an integer  $L = L(v - u)$ , depending only on  $\alpha$  and on the length  $v - u$  — not on the position of  $J$ , nor on  $\beta$  — such that every block of  $L$  consecutive integers  $\{N + 1, \dots, N + L\}$  contains some  $n$  with  $\{n\alpha + \beta\} \in J$ .

We prove this by an explicit *circle walk*, using only Dirichlet's approximation theorem. Put  $\varepsilon := v - u > 0$  and choose an integer  $Q > 2/\varepsilon$ . By Dirichlet's theorem there are integers  $1 \leq n_0 \leq Q$  and  $m$  with  $|n_0\alpha - m| < 1/Q$ . Set

$$\theta := \|n_0\alpha\| = |n_0\alpha - m| \in (0, \frac{\varepsilon}{2}),$$

which is nonzero because  $\alpha$  is irrational. Thus one step of  $n_0$  in the index moves the point  $\{n\alpha + \beta\}$  around the circle by exactly  $\pm\theta$ , the sign being that of  $n_0\alpha - m$  and the same at every step.

Fix any  $N \geq 0$  and consider the  $\lceil 1/\theta \rceil + 1$  indices

$$n_j := N + (j + 1)n_0, \quad j = 0, 1, \dots, \lceil 1/\theta \rceil,$$

all of which lie in the block  $\{N + 1, \dots, N + L\}$ : indeed  $n_0 \geq 1$ , and the largest is  $N + (\lceil 1/\theta \rceil + 1)n_0 = N + L$ . The points  $x_j := \{n_j\alpha + \beta\}$  satisfy  $x_{j+1} = \{x_j \pm \theta\}$ : they march around the circle in steps of the fixed length  $\theta$ , in a fixed direction, wrapping when they pass 0 (resp. 1). In  $\lceil 1/\theta \rceil$  such steps the total displacement is at least 1, so the walk traverses the whole circle. Since each step has length  $\theta < \varepsilon/2 < \varepsilon$ , the walk cannot step over the arc  $J$ , which has length  $\varepsilon$ : some  $x_j$  lies in  $J$ .

Hence every block of

$$L := n_0(\lceil 1/\theta \rceil + 1) \leq Q(\lceil 1/\theta \rceil + 1)$$

consecutive integers contains an index  $n$  with  $\{n\alpha + \beta\} \in J$ . The bound  $L$  depends only on  $\alpha$  and on  $\varepsilon = v - u$  — not on where  $J$  sits, and not on  $\beta$ , since the argument used only the step length  $\theta$  and the direction, both independent of the starting point. Fix this  $L$  for the specific arc used below.

This is the route the formalization takes (**walk\_entrers**); it avoids the three-distance theorem, and indeed any density or equidistribution statement, at the cost of a worse but perfectly explicit constant.

*Step 2: the sub-window claim.* Fix  $\eta \in (0, \frac{1}{2})$ . We claim there is  $S_0 = S_0(\eta, \alpha, \beta, k)$  such that for every  $s \geq S_0$ , the set  $W_s$  contains every integer  $w$  for which

$$\theta := X_s - w \in [\eta, k - \eta]. \quad (10)$$

That is: within the real window  $(X_s - k, X_s]$  of (9), every integer point whose distance  $\theta$  below  $X_s$  lies in  $[\eta, k - \eta]$  is attained.

To see this, fix  $s \geq S_0$  and an integer  $w$  with  $\theta \in [\eta, k - \eta]$ . It suffices to produce pairwise distinct indices  $i_1, \dots, i_k \geq 1$  with

$$\sum_{t=1}^k i_t = s \quad \text{and} \quad \sum_{t=1}^k \{i_t \alpha + \beta\} = \theta \quad \text{exactly,} \quad (11)$$

since then Step 0 gives  $\sum_t q_{i_t} = X_s - \theta = w$ , i.e.  $w \in W_s$ .

Set  $\kappa := \eta/4$  and consider the open real interval

$$\mathcal{I} := (\max(0, \theta - 1) + \kappa, \min(k - 1, \theta) - \kappa).$$

It is nonempty with  $|\mathcal{I}| \geq \eta/2$ , in each of the three regimes of  $\theta$ : if  $\eta \leq \theta \leq 1$  then  $|\mathcal{I}| = \theta - 2\kappa \geq \eta/2$ ; if  $1 < \theta \leq k - 1$  then  $|\mathcal{I}| = 1 - \eta/2 > \eta/2$ ; and if  $k - 1 < \theta \leq k - \eta$  then  $|\mathcal{I}| = (k - \theta) - 2\kappa \geq \eta/2$ .

Pick any  $T^* \in \mathcal{I}$  and set the target arc

$$J := \left( \frac{T^*}{k-1} - \frac{\kappa}{k-1}, \frac{T^*}{k-1} + \frac{\kappa}{k-1} \right) \subseteq (0, 1),$$

of length  $2\kappa/(k-1) > 0$ ; let  $L = L(2\kappa/(k-1))$  be its syndeticity constant from Step 1. Choose the  $k-1$  free indices  $i_1 < \dots < i_{k-1}$ , all with  $\{i_t \alpha + \beta\} \in J$ , from the lowest available blocks: for  $j = 1, \dots, k-1$  the block  $B_j := \{(j-1)L + 1, \dots, jL\}$  contains, by Step 1, at least one index  $i_j$  with  $\{i_j \alpha + \beta\} \in J$ . These lie in disjoint blocks, so they are distinct, and

$$i_j \leq jL, \quad \sum_{t=1}^{k-1} i_t \leq L \frac{k(k-1)}{2}, \quad i_{k-1} \leq (k-1)L. \quad (12)$$

Define the threshold

$$S_0 := 1 + \left\lceil L(k-1) \left( \frac{k}{2} + 1 \right) \right\rceil \quad (13)$$

(the ceiling makes it an integer:  $L$  is an integer, but the factor  $\frac{k}{2}$  is a half-integer when  $k$  is odd), and assume  $s \geq S_0$ . Set the forced last index  $i_k := s - \sum_{t=1}^{k-1} i_t$ . Then by (12) and (13),

$$i_k \geq s - L \frac{k(k-1)}{2} \geq S_0 - L \frac{k(k-1)}{2} = 1 + (k-1)L > (k-1)L \geq i_{k-1},$$

so  $i_k \geq 1$  and  $i_k > i_{k-1} > \dots > i_1$ : the indices  $i_1, \dots, i_k \geq 1$  are pairwise distinct with  $\sum_t i_t = s$ .

With these choices, writing  $F := \sum_{t=1}^{k-1} \{i_t \alpha + \beta\}$ ,

$$F \in (T^* - \kappa, T^* + \kappa) \subseteq (\max(0, \theta - 1), \min(k - 1, \theta)), \quad (14)$$

the last inclusion because  $T^* \in \mathcal{I}$  sits at distance  $> \kappa$  from each endpoint. Now put  $\Theta := \sum_{t \leq k} \{i_t \alpha + \beta\} = F + \{i_k \alpha + \beta\}$ . From (14) and  $\{i_k \alpha + \beta\} \in [0, 1)$ ,

$$\Theta > \max(0, \theta - 1) + 0 \geq \theta - 1, \quad \Theta < \min(k - 1, \theta) + 1 \leq \theta + 1,$$

so  $\Theta \in (\theta - 1, \theta + 1)$ . On the other hand

$$\Theta = \sum_{t \leq k} (i_t \alpha + \beta) - \sum_{t \leq k} [i_t \alpha + \beta] = X_s - \sum_{t \leq k} q_{i_t} \in X_s - \mathbb{Z} = \theta + \mathbb{Z},$$

using  $\sum_{t \leq k} (i_t \alpha + \beta) = \alpha s + k\beta = X_s$  exactly and  $w = X_s - \theta \in \mathbb{Z}$ . The unique element of the coset  $\theta + \mathbb{Z}$  lying in the open interval  $(\theta - 1, \theta + 1)$  is  $\theta$  itself, so  $\Theta = \theta$ . This proves (11) and hence Step 2.

*Step 3: the ladder lands in the sub-window.* Instantiate Step 2 with

$$\eta := \frac{1}{4} \min(\alpha, 1 - \alpha) \in (0, \frac{1}{8}], \quad (15)$$

and fix the resulting  $S_0$ . Let  $x$  be a large integer. Because  $\gcd(\delta, e) = 1$ , the equation  $x = \delta s + ew$  in nonnegative integers admits, for each residue of  $s$ , at most one  $w$ ; the admissible  $s$  are exactly



those with  $s \equiv x\delta^{-1} \pmod{e}$ , an arithmetic progression of common difference  $e$ , and for each we set  $w(s) := (x - \delta s)/e \in \mathbb{Z}$ . Define

$$\theta(s) := X_s - w(s) = (s\alpha + k\beta) - \frac{x - \delta s}{e}.$$

As  $s$  advances by one admissible step ( $s \mapsto s + e$ ),  $X_s$  increases by  $e\alpha$  while  $w(s)$  decreases by  $\delta$ , so

$$\theta(s + e) - \theta(s) = e\alpha + \delta =: \gamma > 0. \quad (16)$$

Thus  $\theta$  is, along the admissible progression, a strictly increasing arithmetic sequence of common difference exactly  $\gamma$ .

The sub-window (10) is the real interval  $\theta \in [\eta, k - \eta]$ , of length  $k - 2\eta$ . Using  $\delta + e = \delta' \leq k$  and (15),

$$\gamma = \delta + e\alpha = (\delta + e) - e(1 - \alpha) = \delta' - e(1 - \alpha) \leq k - (1 - \alpha),$$

since  $e \geq 1$  and  $1 - \alpha > 0$ . As  $2\eta \leq \frac{1}{2}(1 - \alpha) < 1 - \alpha$ , we conclude

$$\gamma \leq k - (1 - \alpha) < k - 2\eta, \quad (17)$$

i.e. the ladder step is *strictly* shorter than the sub-window.

A strictly increasing arithmetic sequence of common difference  $\gamma$  cannot skip over a real interval of length  $> \gamma$ : if  $\theta(s) < \eta$  and  $\theta(s + e) > k - \eta$  for two consecutive admissible terms, then  $\gamma > (k - \eta) - \eta = k - 2\eta$ , contradicting (17). As  $s$  runs through the admissible progression,  $\theta(s)$  increases without bound and is very negative for the smallest admissible  $s$  (where  $w(s) \approx x/e$  dominates), so it crosses  $[\eta, k - \eta]$ ; by the no-skip property some admissible  $s^*$  has  $\theta(s^*) \in [\eta, k - \eta]$ .

It remains to say what “ $x$  large” means, and we make the two thresholds explicit. Solving  $\theta(s) = \theta$  for  $s$  gives  $s\gamma = x + e\theta - ek\beta$ , so any  $s^*$  with  $\theta(s^*) \in [\eta, k - \eta] \subseteq [0, k]$  satisfies

$$\frac{x - ek|\beta|}{\gamma} \leq s^* \leq \frac{x + ek(1 + |\beta|)}{\gamma}. \quad (18)$$

Two conditions are needed.

- $s^* \geq S_0$ , so that Step 2 applies. By the left inequality of (18) and  $\gamma \leq k$ , it suffices that

$$x \geq kS_0 + ek|\beta|.$$

- $w(s^*) \geq 0$ , so that the representation  $x = \delta s^* + ew(s^*)$  uses a nonnegative  $w$ . This says  $\delta s^* \leq x$ . By the right inequality of (18) this holds as soon as  $\delta(x + ek(1 + |\beta|)) \leq \gamma x = (\delta + e\alpha)x$ , i.e. as soon as

$$x \geq \frac{\delta k(1 + |\beta|)}{\alpha},$$

the denominator being nonzero because  $\alpha \in (0, 1)$ .

So the conclusion holds for every integer

$$x \geq X_0 := 1 + \max\left(kS_0 + ek|\beta|, \frac{\delta k(1 + |\beta|)}{\alpha}\right),$$

an explicit threshold depending only on  $k, \delta, e, \alpha, \beta$  (through  $S_0$ , which was fixed in Step 2). Step 2 then gives  $w(s^*) \in W_{s^*}$ , whence

$$x = \delta s^* + ew(s^*) \in kA - ka_1.$$

As  $x$  was an arbitrary large integer,  $kA - ka_1$  contains all large integers; in the original, unscaled coordinates this is a full congruence-class tail modulo  $g$ , so  $A$  is tail-covering.  $\square$

**Theorem 3.12** (Two-letter core). *Let  $k \geq 3$  and  $d_2 \leq k$ . Every  $(d_1, d_2)$ -sequence whose tail alphabet has at most two letters is tail-covering. In particular, with Lemma 3.1,  $r_k(d_1, d_1 + 1)$  does not exist whenever  $d_1 + 1 \leq k$ , in the strong form.*

*Proof.* Lemmas 3.3, 3.4, 3.5, 3.8 (with Lemma 3.10 at the even boundary) and 3.11 exhaust the cases.  $\square$

#### 3.4. Three or more letters: widths are not the obstacle.

*Remark 3.13.* The antidiagonal width  $W_2$  of §3.3 was defined through the binary counting function  $q_n$ , and has no meaning for an alphabet of three or more letters. We do not attempt to generalize it, because the route below does not need it. Indeed Lemma 3.6 already *fails* for three or more letters: configuration moves change values by letter *differences*, which can exceed 1, so  $W_s$  is dense in its span but may have holes, and Lemma 3.7 does not apply. The slot lemma below bypasses the antidiagonal analysis entirely.

#### 3.5. The slot lemma.

**Lemma 3.14** (Slot lemma). *Let  $A$  have tail alphabet  $G_\infty$  with  $|G_\infty| \geq 3$ ,  $\gcd(G_\infty) = 1$  (after Lemma 3.4), and  $M := \max G_\infty \leq d_2$ . If  $k - 1 \geq m(G_\infty)$ , then  $A$  is tail-covering.*

*Proof.* Let  $S$  realize  $m(G_\infty)$ : a multiset with  $|S| = m \leq k - 1$ , drawn from  $G_\infty$ , whose subset sums contain  $\{c, c + 1, \dots, c + M - 1\}$ . Enumerate  $S = \{\delta_1, \dots, \delta_m\}$  (values in  $G_\infty$ , repeats allowed). Because each value of  $G_\infty$  occurs infinitely often in the gap sequence, we may pick  $m$  distinct, pairwise non-adjacent positions  $n_1, \dots, n_m$  with  $g_{n_t} = \delta_t$ .

Call  $t$  a *fine slot*: a summand index  $i_t \in \{n_t - 1, n_t\}$ , contributing  $p_{n_t-1}$  (“off”) or  $p_{n_t-1} + \delta_t$  (“on”). Over all on/off choices the  $m$  fine slots contribute

$$\text{base} + \text{subset-sums}(S) \supseteq \text{base} + [c, c + M - 1], \quad \text{base} := \sum_t p_{n_t-1}.$$

Pack the remaining  $k - 1 - m$  summands at fixed indices, folding their contribution into base (repeated indices are allowed in  $kA$ ). This uses  $k - 1$  of the  $k$  summands and realizes every value of an interval of  $M$  consecutive integers.

Two endpoint facts should be made explicit. First, each value of  $G_\infty$  recurs infinitely often, so the positions  $n_t$  may be taken as far out as we please; we take them all beyond the tail index  $T$  of Lemma 3.2, so that  $n_t - 1 \geq T$  and the “off” choice  $p_{n_t-1}$  is a legitimate prefix sum of the tail.

Second, the coarse dial. In the tail, consecutive prefix sums differ by  $g_{n+1} \in G_\infty$ , hence by at least 1 and at most  $M$ ; so  $n \mapsto p_n$  is strictly increasing with steps of size at most  $M$ , and  $p_n \rightarrow \infty$ . Fix  $x$  and put  $\phi(n) := x - \text{base} - p_n$ , which is strictly decreasing with steps of size at most  $M$  and tends to  $-\infty$ . Take  $x$  large enough that  $\phi(T) > c + M - 1$  *strictly* — possible because base,  $T$  and  $c$  are fixed while  $x$  grows. Then  $\phi$  starts strictly above the target interval  $[c, c + M - 1]$  and eventually falls below it, so there is a *first* index  $n^* > T$  with  $\phi(n^*) \leq c + M - 1$ ; note  $n^* > T$ , so  $n^* - 1 \geq T$  is a legitimate tail index. By minimality  $\phi(n^* - 1) > c + M - 1$ , and since  $\phi$  decreases by at most  $M$  in one step,

$$\phi(n^*) > (c + M - 1) - M = c - 1, \quad \text{i.e.} \quad \phi(n^*) \geq c.$$

Hence  $\phi(n^*) \in [c, c + M - 1]$ , an interval realized by the fine slots. This first-crossing argument is exactly where a run of  $M$  consecutive integers is needed: a run of length  $M - 1$  would not suffice, since the dial can step by  $M$ .

Therefore  $x \in kA - ka_1$ , and  $kA - ka_1$  contains all large integers (in rescaled coordinates; a full class tail modulo  $g$  in general).  $\square$

The binding corner is  $k = d_2 = M$  — the adversary uses the maximal gap infinitely often and  $k$  is as small as the hypothesis  $d_2 \leq k$  allows. There the requirement  $k - 1 \geq m(G_\infty)$  becomes *exactly*  $m(G_\infty) \leq M - 1$ , which is precisely Theorem B. Granting it:  $k - 1 \geq d_2 - 1 \geq M - 1 \geq m(G_\infty)$  for all  $k \geq d_2$  (and for rescaled alphabets  $M/g - 1 < d_2 - 1$  a fortiori).

For perspective we record a short self-contained bound in the same direction. It is not needed below, but it exhibits the mechanism. It is genuinely weaker than Theorem B: besides the worse bound, its hypothesis can fail — for  $G = \{6, 10, 15\}$ , of gcd 1, no element is coprime to  $\min G = 6$ .

**Lemma 3.15** (Base grid). *Let  $G$  satisfy  $|G| \geq 3$ ,  $\gcd G = 1$ ,  $\min G = a$ ,  $\max G = M$ , and suppose  $G$  contains an element  $c$  with  $\gcd(a, c) = 1$ . Then  $m(G) \leq a + M - 1 \leq 2M - 3$ .*

*Proof.* Take  $a - 1$  copies of  $c$  and  $p := \lceil ((a - 1)c + M - 1)/a \rceil$  copies of  $a$ . For  $n \in [(a - 1)c, pa]$ , the unique  $i \in [0, a - 1]$  with  $ic \equiv n \pmod{a}$  satisfies  $ic \leq (a - 1)c \leq n$  and  $(n - ic)/a \in [0, p]$ , so the subset sums cover that interval, whose length is  $\geq M$ . Since  $p \leq M$ , the size is  $\leq a + M - 1$ ; and  $|G| \geq 3$  forces  $a \leq M - 2$ .  $\square$

### 3.6. Conclusion of the non-existence half.

**Proposition 3.16.** *Modulo Theorem B, every  $(d_1, d_2)$ -sequence with  $d_2 \leq k$  and  $k \geq 3$  is tail-covering.*

*Proof.* Tails with at most two letters are covered by Theorem 3.12. Tails with three or more letters are covered by Lemma 3.14 together with Theorem B, which supplies  $m(G_\infty) \leq \max(G_\infty) - 1 \leq d_2 - 1 \leq k - 1$ .  $\square$

With Lemma 3.1 this proves the non-existence half of Theorem A, in the strong form. It remains to prove Theorem B.

## 4. THE BOUNDED SUBSET-SUM INTERVAL THEOREM

We restate the target.

**Theorem 4.1** (= Theorem B). *For every finite  $G \subset \mathbb{Z}_{\geq 1}$  with  $|G| \geq 3$ ,  $\gcd(G) = 1$  and  $\max(G) = M$ , there is a multiset  $S$  of elements of  $G$  with  $|S| \leq M - 1$  whose subset sums contain  $M$  consecutive integers.*

If  $a := \min G = 1$ , take  $M - 1$  copies of 1. So assume  $a \geq 2$ ; the hard core below will force  $a \geq 3$ .

### 4.1. Machinery.

**Lemma 4.2** (Two-generator interval). *Let  $\alpha < \beta$  be coprime positive integers, and let  $x \geq \beta - 1$ ,  $y \geq \alpha - 1$ . Then, with  $C := (\alpha - 1)(\beta - 1)$ ,*

$$\{i\alpha + j\beta : 0 \leq i \leq x, 0 \leq j \leq y\} \supseteq [C, \alpha x + \beta y - C].$$

*Proof.* If  $\alpha = 1$  then  $C = 0$ ; for  $n \in [0, \alpha x + \beta y]$  take  $j := \min(y, \lfloor n/\beta \rfloor)$ , so that  $0 \leq n - j\beta \leq \max(\beta - 1, n - y\beta) \leq x$ .

Let  $\alpha \geq 2$ . *Base case  $y = \alpha - 1$ :* for  $n \in [C, \alpha x + \beta(\alpha - 1) - C]$  pick the unique  $j \in [0, \alpha - 1]$  with  $j\beta \equiv n \pmod{\alpha}$ . Then  $n - j\beta \geq C - (\alpha - 1)\beta = -(\alpha - 1)$  and is  $\equiv 0 \pmod{\alpha}$ , hence  $\geq 0$ ; and  $i := (n - j\beta)/\alpha \leq (\alpha x + (\alpha - 1)\beta - C)/\alpha \leq x + (\alpha - 1)/\alpha$ , so  $i \leq x$ .

*Step  $y \rightarrow y + 1$ :* the new part of the interval is  $n \in (\alpha x + \beta y - C, \alpha x + \beta(y + 1) - C]$ . Then  $n - \beta \in [C, \alpha x + \beta y - C]$  — the lower end because  $\alpha x + \beta y \geq 2C + \beta$  for  $x \geq \beta - 1$ ,  $y \geq \alpha - 1$ ,  $\alpha \geq 2$  — so by induction  $n - \beta = i\alpha + j\beta$  with  $i \leq x$ ,  $j \leq y$ , and thus  $n = i\alpha + (j + 1)\beta$ .  $\square$

**Lemma 4.3** (Frame lemma). *Let  $g_1, g_2, \nu$  be positive integers and  $L \geq 1$ . Suppose that for every residue  $\rho \in \mathbb{Z}_\nu$  there is a pair  $(j_\rho, k_\rho)$  with  $0 \leq j_\rho \leq Y$ ,  $0 \leq k_\rho \leq Z$  and  $j_\rho g_1 + k_\rho g_2 \equiv \rho \pmod{\nu}$ . Put  $S := \max_\rho (j_\rho g_1 + k_\rho g_2)$  and  $x := \lceil (L-1+S)/\nu \rceil$ . Then the multiset consisting of  $Y$  copies of  $g_1$ ,  $Z$  copies of  $g_2$  and  $x$  copies of  $\nu$  — of size  $Y+Z+x$  — has subset sums containing every integer of  $[S, \nu x] \supseteq [S, S+L-1]$ .*

*Proof.* For  $n \in [S, \nu x]$  let  $\rho := n \bmod \nu$  and  $r := j_\rho g_1 + k_\rho g_2 \leq S \leq n$ . Then  $t := (n-r)/\nu$  is a nonnegative integer  $\leq x$ , and taking  $j_\rho$  copies of  $g_1$ ,  $k_\rho$  of  $g_2$  and  $t$  of  $\nu$  realizes  $n$ .  $\square$

In all uses below  $\nu = a \in G$ ,  $(g_1, g_2) = (b, M)$  and  $L = M$ ; the *budget* of a box  $(Y, Z)$  is

$$B = \left\lceil \frac{M-1+S}{a} \right\rceil + Y + Z,$$

with  $S$  the maximum, over residue classes, of the minimal representative height in the box.

**Lemma 4.4** (Staircase windows and merges). *Let  $G = \{a, b, M\}$  with  $\gcd(a, b) = 1$ , and set  $e := b - a$ ,  $\mu := M - a$ ,  $g := \gcd(e, \mu)$ ,  $e' := e/g$ ,  $\mu' := \mu/g$  (coprime, with  $\mu' \geq 2$  since  $h := M - b \geq 1$ ), and  $C' := (e' - 1)(\mu' - 1)$ . Note that  $\gcd(a, b) = 1$  forces  $\gcd(a, g) = 1$ : a common prime of  $g$  and  $a$  would divide  $e$  and  $a$ , hence  $b$ . Fix a multiset  $(x, y, z)$  of copies of  $a, b, M$  with  $y \geq \mu' - 1$  and  $z \geq e' - 1$ , and put  $c := y + z$  and  $V' := y e' + z \mu' - C'$ . For  $0 \leq t \leq x + y + z$  the subset sums using exactly  $t$  elements form  $ta + \mathcal{O}_t$ , where  $\mathcal{O}_t = \{je + k\mu : j \leq y, k \leq z, \max(0, t-x) \leq j+k \leq t\}$ , and:*

- (a) if  $c \leq t \leq x$  then  $\mathcal{O}_t \supseteq g[C', V']$ ;
- (b) (two-frame merge,  $g = 1$ ) if  $x = c + 1$  and  $V - C \geq a - 1$  (writing  $V = V'$ ,  $C = C'$ ), the sums cover  $[ca + C, (c+1)a + V]$ , of length  $a + (V - C) + 1$ ;
- (c) (short two-frame merge,  $g = 1$ ) if  $x = c$ , then level  $c + 1$  imposes only  $j + k \geq 1$ , which excludes exactly the offset 0; so  $\mathcal{O}_{c+1} \supseteq [\max(C, 1), V]$ , and if  $V \geq a + \max(C, 1) - 1$  the sums cover  $[ca + C, (c+1)a + V]$ , of length  $a + V - C + 1$ ;
- (d) ( $g$ -phase union,  $g \geq 2$ ) if  $x = c + g$  then all levels  $t \in [c, c+g]$  satisfy (a); since  $\gcd(a, g) = 1$ , each residue class modulo  $g$  of the target is served by a unique  $t_0 \in [c, c+g-1]$ , and the sums cover the solid interval  $[(c+g-1)a + gC', ca + gV']$ , of length  $g(V' - C') - (g-1)a + 1$  (the base form: one frame per class, no further hypothesis needed). More generally, if

$$x \geq c + g \quad \text{and} \quad V' - C' \geq a - 1 \quad (\text{the frame-merge condition}),$$

then within each residue class modulo  $g$  the frames  $t_0, t_0 + g, t_0 + 2g, \dots \leq x$  merge, and the sums cover the solid interval  $[(c+g-1)a + gC', (x-g+1)a + gV']$  (the extended form).

*Proof.* (a) For  $t \in [c, x]$  every  $(j, k)$  in the box has  $j + k \leq c \leq t$  and  $i = t - j - k \in [0, x]$ ; apply Lemma 4.2 to  $(e', \mu')$  and scale by  $g$ .

(b) Levels  $c$  and  $c+1$  cover  $ta + [C, V]$  each; they merge exactly when  $ca + V \geq (c+1)a + C - 1$ .

(c) Every nonzero offset of the box keeps all of its representations at level  $c + 1$ , each having  $j + k \geq 1$ .

(d), *base form.* Every  $n$  in the displayed interval satisfies  $n \equiv t_0 a \pmod{g}$  for a unique  $t_0 \in [c, c+g-1]$ , and  $u := (n - t_0 a)/g$  satisfies  $u \geq C'$  (from the left endpoint, since  $t_0 \leq c+g-1$ ) and  $u \leq V'$  (from the right endpoint, since  $t_0 \geq c$ ). By (a), level  $t_0$  realizes  $t_0 a + gu = n$ .

(d), *extended form.* Fix a residue class modulo  $g$  and its unique  $t_0 \in [c, c+g-1]$ . The levels serving this class within  $[c, x]$  are exactly the frames  $t_j := t_0 + jg$  for  $0 \leq j \leq J$ ,  $J := \lfloor (x - t_0)/g \rfloor \geq 0$  (well defined since  $t_0 \leq c+g-1 \leq x-1 < x$ , using  $x \geq c+g$ ). By (a), frame  $t_j$  covers  $t_j a + g[C', V']$ , an arithmetic progression of step  $g$  inside the class  $t_0 a + g\mathbb{Z}$ . Two consecutive frames  $t_j, t_{j+1}$  leave no gap of the class between them iff  $t_{j+1}a + gC' \leq t_j a + gV' + g$ , i.e. iff  $ga + gC' \leq gV' + g$ , i.e. iff  $V' - C' \geq a - 1$  — precisely the frame-merge condition, under

which the class is covered solidly (in steps of  $g$ ) on  $[t_0a + gC', (t_0 + Jg)a + gV']$ . When  $J = 0$  this holds trivially. Over  $t_0 \in [c, c + g - 1]$  the left endpoints are all  $\leq (c + g - 1)a + gC'$ ; and by maximality of  $J$  we have  $t_0 + (J + 1)g > x$ , i.e.  $t_0 + Jg > x - g$ , so the right endpoints are all  $\geq (x - g + 1)a + gV'$ . Hence every integer of  $[(c + g - 1)a + gC', (x - g + 1)a + gV']$  lies in the solidly covered range of its own class.  $\square$

*Remark 4.5* (The merge condition is not removable from the extended form). For  $(a, b, M) = (5, 7, 9)$  — so  $e = 2$ ,  $\mu = 4$ ,  $g = 2$ ,  $e' = 1$ ,  $\mu' = 2$ ,  $C' = 0$  — take  $y = 1$ ,  $z = 0$  (hence  $c = 1$ ,  $V' = 1$ ) and  $x = 3 = c + g$ . The merge condition fails ( $V' - C' = 1 < 4 = a - 1$ ), and so does the conclusion: the multiset  $\{5, 5, 5, 7\}$  has subset sums  $\{0, 5, 7, 10, 12, 15, 17, 22\}$ , which miss 11 inside the would-be interval  $[10, 12]$ . An earlier form of Lemma 4.4(d) omitted this hypothesis and was false; the counterexample is pinned as a decided guard in the formalization.

**Lemma 4.6** ( $\lambda$ -lift). *Suppose the box  $(Y, Z)$  modulo  $a$  (Lemma 4.3 with  $\nu = a$ ,  $(g_1, g_2) = (b, M)$ ) covers all residues with budget  $B \leq M - 1$  for the triple  $(a, b, M)$ , and  $Y + Z \leq a - 1$ . Then the same box has budget  $\leq M' - 1$  for the triple  $(a, b + a, M + a)$ , where  $M' := M + a$ .*

*Proof.* Residues of  $jb + kM$  modulo  $a$  are unchanged, so the same representative pairs serve. Each representative height grows by  $(j + k)a \leq (Y + Z)a$ , so  $S' \leq S + (Y + Z)a$  and

$$\left\lceil \frac{M' - 1 + S'}{a} \right\rceil \leq \left\lceil \frac{M - 1 + S}{a} \right\rceil + Y + Z + 1.$$

Thus the budget grows by at most  $Y + Z + 1 \leq a$ , while the target  $M - 1$  grows by exactly  $a$ .  $\square$

## 4.2. Reduction to the hard core.

**Lemma 4.7** (Graham reduction). *Theorem 4.1 for all alphabets follows from Theorem 4.1 for irreducible ones: those with  $|G| = 3$ , or with  $|G| \geq 4$  and  $G$  minimal (no proper subset of size  $\geq 3$  has  $\gcd 1$ ).*

*Proof.* We use Graham's growth lemma [Gr64]: if the subset sums of a multiset contain a run of  $\ell$  consecutive integers and  $g \in G$  satisfies  $g \leq \ell$ , then adjoining one copy of  $g$  extends the run to length  $\ell + g$ .

Induct on  $|G|$ . If  $|G| \geq 4$  is not irreducible, some  $e \in G$  is redundant, i.e.  $\gcd(G \setminus \{e\}) = 1$  and  $|G \setminus \{e\}| \geq 3$ . If  $e \neq M$ , any witness for  $G \setminus \{e\}$  is a witness for  $G$ . If  $e = M$ , let  $M' := \max(G \setminus \{M\})$ ; by induction a run of length  $\geq M'$  costs  $\leq M' - 1$ , and Graham-filling with copies of  $a = \min G$  costs  $\leq \lceil (M - M')/a \rceil \leq M - M'$ , for a total of  $\leq M - 1$ .  $\square$

**Lemma 4.8** (Small coprime pair). *If some coprime pair  $\alpha, \gamma \in G$  has  $\alpha + \gamma \leq M - 1$ , then  $m(G) \leq M - 1$ .*

*Proof.* Take  $x = M - \alpha$  copies of  $\alpha$  and  $\alpha - 1$  copies of  $\gamma$ . By Lemma 4.3 (modulus  $\alpha$ , representatives  $j\gamma$ ) the sums cover  $[(\alpha - 1)\gamma, x\alpha]$ , of length  $x\alpha - (\alpha - 1)\gamma + 1 \geq (M - \alpha)\alpha - (\alpha - 1)(M - \alpha - 1) + 1 = M$ . The budget is  $(M - \alpha) + (\alpha - 1) = M - 1$ .  $\square$

**Lemma 4.9** (Pair at the boundary). *A coprime pair  $\alpha < \beta$  in  $G$  with  $\alpha + \beta \in \{M, M + 1\}$  gives  $m(G) \leq M - 1$ .*

*Proof.* For an integer  $t \geq 0$ , take  $x = \beta - 1$  copies of  $\alpha$  and  $y = \alpha - 1 + t$  copies of  $\beta$ . The hypotheses of Lemma 4.2 hold, so the subset sums contain the interval  $[C, \alpha x + \beta y - C]$  with  $C = (\alpha - 1)(\beta - 1)$ , of length  $\alpha x + \beta y - 2C + 1 = \alpha + \beta - 1 + t\beta$ , at budget  $x + y = \alpha + \beta - 2 + t$ . The two boundary values need *different*  $t$ :

- If  $\alpha + \beta = M + 1$ , take  $t = 0$ : run length  $\alpha + \beta - 1 = M$ , budget  $\alpha + \beta - 2 = M - 1$ .

- If  $\alpha + \beta = M$ , take  $t = 1$ : run length  $\alpha + \beta - 1 + \beta = M - 1 + \beta \geq M$  (as  $\beta \geq 2$ ), budget  $\alpha + \beta - 2 + 1 = M - 1$ . Here  $t = 0$  would give a run of length only  $M - 1$ ; one further copy of  $\beta$  is needed, and the budget absorbs it exactly.

□

*Example 4.10.* For  $(\alpha, \beta, M) = (2, 3, 5)$  (the case  $\alpha + \beta = M$ ),  $t = 0$  gives the multiset  $\{2, 2, 3\}$  with subset sums  $\{0, 2, 3, 4, 5, 7\}$ : longest run  $[2, 5]$ , of length  $4 < 5$  — insufficient, exactly as the boundary bookkeeping predicts. With  $t = 1$  the multiset is  $\{2, 2, 3, 3\}$ , of budget  $4 = M - 1$ , with subset sums  $\{0, 2, 3, 4, 5, 6, 7, 8, 10\} \supseteq [2, 8]$ : a run of  $7 \geq 5$  consecutive integers.

**Lemma 4.11** (Non-coprime pair). *If  $|G| = 3$  and  $a, b \in G \setminus \{M\}$  have  $d := \gcd(a, b) \geq 2$  — in which case  $\gcd(d, M) = 1$  automatically — then  $m(G) \leq M - 1$ .*

*Proof.* Write  $a = d\alpha$ ,  $b = d\beta$ . Take  $x = \beta - 1$  copies of  $a$ ,  $y = M - d - \beta + 1$  copies of  $b$ , and  $d - 1$  copies of  $M$ : the budget is exactly  $M - 1$ . The  $(a, b)$ -grid gives, by Lemma 4.2 scaled by  $d$ , an arithmetic progression of step  $d$  and length  $L = \alpha x + \beta y - 2(\alpha - 1)(\beta - 1) + 1$ . The requirement  $y \geq \alpha - 1$  follows from  $M \geq d\beta + 1$  via  $(d - 1)(\beta - 1) \geq \alpha - 2$ . Moreover  $L \geq M + 1$ : this is monotone in  $M$ , and at the worst case  $M = d\beta + 1$  it reduces to  $d\beta(\beta - 2) - \beta^2 - \alpha\beta + \alpha + 4\beta - 3 \geq 0$ , which for  $d \geq 2$  follows from  $\beta(\beta - \alpha) + \alpha - 3 \geq 0$ , true for all  $1 \leq \alpha < \beta$ . Bridging the  $d$  residues by  $kM$  with  $k \leq d - 1$  (using  $\gcd(d, M) = 1$ ) covers  $[dC_{\alpha\beta} + (d - 1)M, dC_{\alpha\beta} + (L - 1)d]$ , of length  $\geq M$ ; the requirement  $(L - 1)d \geq dM - 1$  is equivalent, both sides being considered modulo  $d$ , to  $L \geq M + 1$ . □

**Lemma 4.12** (Structure of minimal alphabets). *Let  $G = \{g_1, \dots, g_m\}$  be minimal with  $m \geq 4$ , and put  $d_i := \gcd(G \setminus \{g_i\})$ . Then each  $d_i \geq 2$ ; the  $d_i$  are pairwise coprime; and  $d_j \mid g_i$  for every  $j \neq i$ . Hence every  $g_i$  is divisible by  $\prod_{j \neq i} d_j$ , a product of  $m - 1 \geq 3$  pairwise coprime integers  $\geq 2$ , so  $g_i \geq 2 \cdot 3 \cdot 5 = 30$  when  $m = 4$ , and  $g_i \geq 210$  when  $m \geq 5$ . Also every pair of elements of  $G$  has  $\gcd \geq 2$ .*

*Proof.* Each  $d_i \geq 2$  by minimality. For  $i \neq j$ ,  $\gcd(d_i, d_j)$  divides every element of  $(G \setminus \{g_i\}) \cup (G \setminus \{g_j\}) = G$ , hence divides  $\gcd(G) = 1$ ; so the  $d_i$  are pairwise coprime. That  $d_j \mid g_i$  for  $j \neq i$  is immediate from the definition. Finally a coprime pair would lie in a gcd-1 proper triple, contradicting minimality. □

There are exactly 6 minimal 4-alphabets with  $M \leq 110$  and 12 with  $M \leq 130$ , the smallest being  $\{30, 42, 70, 105\}$  with  $M = 105$ ; no minimal alphabet of any size has  $M < 105$ , and  $|G| \geq 5$  is impossible below  $M = 210$ . (This enumeration is machine-checked two independent ways; see Appendix A.)

**Lemma 4.13** (Minimal alphabets,  $|G| \geq 4$ ). *If  $G$  is minimal with  $|G| \geq 4$ , then  $m(G) \leq M - 1$ .*

*Proof.* Remove the smallest element  $a$ . Then  $G' := G \setminus \{a\}$  has  $\gcd(G') =: \delta \geq 2$  by minimality, contains  $M$ , and  $H := G'/\delta$  is an alphabet with  $\gcd(H) = 1$ ,  $\max H = M/\delta < M$  and  $|H| \geq 3$ . By strong induction on  $M$ , a run of length  $\max H$  costs  $\leq M/\delta - 1$  within  $H$ . Graham-filling with  $a' := \min H$  up to length  $\ell := \lceil (M - 1 + (\delta - 1)a)/\delta \rceil + 1$  costs  $\leq (M/\delta - 1) + \lceil (\ell - M/\delta)/a' \rceil$ . Scaling by  $\delta$  and bridging the  $\delta$  residues with  $k \leq \delta - 1$  copies of  $a$  (legitimate since  $\gcd(a, \delta) = 1$ , because  $\gcd(G) = 1$ ) covers an interval of length  $\geq M$  at total budget

$$T \leq \left(\frac{M}{\delta} - 1\right) + \left\lceil \frac{\ell - M/\delta}{a'} \right\rceil + (\delta - 1).$$

Now  $a < \min G' = \delta a'$ , i.e.  $a \leq \delta a' - 1$ , whence  $\ell - M/\delta \leq (\delta - 1)a/\delta + 2 < (\delta - 1)a' + 2$  and  $\lceil (\ell - M/\delta)/a' \rceil \leq \delta + 1$ . So  $T \leq M/\delta + 2\delta - 1$ , and

$$T \leq M - 1 \iff M\left(1 - \frac{1}{\delta}\right) \geq 2\delta \iff M \geq \frac{2\delta^2}{\delta - 1}.$$



For  $\delta \geq 3$  we have  $2\delta^2/(\delta-1) \leq 3\delta \leq M$ , since  $G'$  contains at least three distinct multiples of  $\delta$ . For  $\delta = 2$  the requirement is  $M \geq 8$ , which holds since every element is  $\geq 30$  by Lemma 4.12. The bound never divides by  $a'$  except through  $a \leq \delta a' - 1$ , so the case  $a' = 1$  is subsumed.  $\square$

**4.3. The hard core.** By Lemmas 4.7–4.13, Theorem 4.1 reduces to triples  $G = \{a, b, M\}$  with  $\gcd(a, b) = 1$  and — applying Lemmas 4.8 and 4.9 to the pair  $(a, b)$  — with  $\delta := a + b - M \geq 2$ . We fix the following notation for the rest of the section:

$e := b - a \geq 1$ ,  $h := M - b \in [1, a - 2]$ ,  $\mu := e + h = M - a$ ,  $\delta = a - h$ ,  $\gcd(a, e) = 1$ ,  $a \geq 3$ ,  
 $g := \gcd(e, h) = \gcd(e, \mu)$  (coprime to each of  $a, b, M$ ),  $\bar{e} := e \bmod a \in [1, a - 1]$ ,  $\bar{\mu} := M \bmod a$ ,  
 and — when  $\bar{\mu} \neq 0$  —  $\eta := \bar{\mu} \bar{e}^{-1} \bmod a$ .

These symbols recur densely for the rest of the section, so we collect them.

| Symbol      | Meaning                                | Symbol      | Meaning                                |
|-------------|----------------------------------------|-------------|----------------------------------------|
| $a < b < M$ | the hard-core triple, $\gcd(a, b) = 1$ | $g$         | $\gcd(e, h) = \gcd(e, \mu)$            |
| $e$         | $b - a \geq 1$                         | $e', \mu'$  | $e/g, \mu/g$ (coprime)                 |
| $h$         | $M - b \in [1, a - 2]$                 | $C'$        | $(e' - 1)(\mu' - 1)$                   |
| $\mu$       | $e + h = M - a$                        | $V'$        | $ye' + z\mu' - C'$                     |
| $\delta$    | $a + b - M = a - h \geq 2$             | $(x, y, z)$ | copies of $a, b, M$                    |
| $\bar{e}$   | $e \bmod a \in [1, a - 1]$             | $c$         | $y + z$                                |
| $\bar{\mu}$ | $M \bmod a$                            | $B$         | budget = $x + y + z$                   |
| $\eta$      | $\bar{\mu} \bar{e}^{-1} \bmod a$       | $\lambda$   | $\lfloor e/a \rfloor$ (the lift index) |

Note that  $\eta = 1$  is impossible ( $\eta = 1 \iff a \mid h$ , but  $1 \leq h \leq a - 2$ ), and that

$$\eta = 0 \iff a \mid M, \quad \eta = a - 1 \iff a \mid b + M,$$

these two lines being disjoint (both would force  $a \mid b$ ). The target budget is  $M - 1$  throughout.

Each hard-core triple is routed to exactly one of six branches, evaluated top to bottom:

- **Case D** —  $a \mid M$  (§4.4);
- **Case P** — else  $a \mid b + M$ , the  $\eta = -1$  line (§4.5);
- **Case L** — else  $e = h$ , the scaled consecutive triples (§4.6);
- **Case E** — else  $a \geq 12$  and  $\mu \geq 12$ , the universal  $\eta$ -box (§4.7);
- **Case T** — else  $\mu \leq 11$ , finitely many staircase lines (§4.8);
- **Case B** — else  $a \leq 11$  and  $\mu \geq 12$ , finite bases plus the  $\lambda$ -lift (§4.9).

The branch *conditions* can overlap — e.g.  $(3, 4, 5)$  satisfies both P and L — but they are evaluated as an ordered if–else tree, so the routing assigns every hard-core triple to exactly one branch; exhaustiveness is checked in §4.10. Each yields a covering multiset of size  $\leq M - 1$ . The following table summarizes the tree; “symbolic” means the branch is closed by a closed-form construction valid for all parameters in its range, and “finite” means it additionally consumes a certificate table.

| Branch | Condition                | Construction        | Budget mechanism                                                          | Closure             |
|--------|--------------------------|---------------------|---------------------------------------------------------------------------|---------------------|
| D      | $a \mid M$               | Lemma 4.14          | $(a-1)$ copies of $b$ span $\mathbb{Z}_a$ ;<br>$a, M$ pad at “half price” | symbolic            |
| P      | $a \mid b + M$           | Lemma 4.15          | $(b, M)$ pairs are residue-neutral<br>movers worth $ra$                   | symbolic            |
| L      | $e = h$                  | Lemma 4.19          | staircase, extended form                                                  | symbolic            |
| E      | $a, \mu \geq 12$         | Lemma 4.20          | the $\eta$ -box $(t-1, Z)$                                                | symbolic            |
| T      | $\mu \leq 11$            | Lemmas 4.21, 4.23   | staircase + linear tail;<br>172 exceptions                                | finite<br>(Table A) |
| B      | $a \leq 11, \mu \geq 12$ | Table B + Lemma 4.6 | finite bases, lifted in $\lambda$                                         | finite<br>(Table B) |

Four of the six branches are closed symbolically; only T and B consume finite data, and between them they account for the whole of the finite layer (Proposition A.1).

#### 4.4. Case D: $a \mid M$ .

**Lemma 4.14.** *If  $a \mid M$ , write  $M = qa$  with  $q \geq 2$  and put  $x_{\text{eff}} := \lceil (M - 1 + (a - 1)b)/a \rceil$ . Then the multiset with  $a - 1$  copies of  $b$ ,  $q - 1$  copies of  $a$ , and  $z := \lceil (x_{\text{eff}} - (q - 1))/q \rceil$  copies of  $M$  has budget  $\leq M - 1$ , and its subset sums contain  $M$  consecutive integers.*

*Proof. Coverage.* The  $a - 1$  copies of  $b$  realize  $\{jb : 0 \leq j \leq a - 1\}$ , a complete residue system modulo  $a$  (as  $\gcd(a, b) = 1$ ), with top  $S = (a - 1)b$ . The padding realizes  $\{ia + kM : 0 \leq i \leq q - 1, 0 \leq k \leq z\} = a\{i + qk\} \supseteq a[0, (q - 1) + qz]$ , because the blocks  $[qk, qk + (q - 1)]$  abut, each of length  $q$ ; and  $(q - 1) + qz \geq x_{\text{eff}}$  by the choice of  $z$ . By Lemma 4.3 the subset sums cover  $[S, S + M - 1]$ .

*Budget.* Since  $b \leq M - 1$ ,

$$\frac{M - 1 + (a - 1)b}{a} \leq \frac{M - 1 + (a - 1)(M - 1)}{a} = M - 1,$$

so  $x_{\text{eff}} \leq M - 1 = qa - 1$ , whence

$$z = \left\lceil \frac{x_{\text{eff}} - (q - 1)}{q} \right\rceil \leq \left\lceil \frac{qa - 1 - (q - 1)}{q} \right\rceil = \left\lceil \frac{q(a - 1)}{q} \right\rceil = a - 1.$$

Therefore

$$(a - 1) + (q - 1) + z \leq (a - 1) + (q - 1) + (a - 1) = 2a + q - 3 \leq qa - 1 = M - 1,$$

the last inequality being exactly  $(a - 1)(q - 2) \geq 0$ , valid for all  $q \geq 2$  and  $a \geq 3$ . A single construction therefore suffices for the whole case; no split on  $q$  is needed.  $\square$

**4.5. Case P:  $a \mid b + M$ .** Here  $b + M = ra$  for an integer  $r \geq 3$ , and copies of  $b$  and  $M$  act as opposite residue steps modulo  $a$ : a  $(b, M)$ -pair is a residue-neutral mover worth  $r$  grid units of  $a$  at a cost of 2.

**Lemma 4.15** (Uniform pair-frame). *Let  $(a, b, M)$  be a hard-core triple with  $b + M = ra$ ,  $r \geq 3$ . Then the multiset*

$$(x, y, z) = \left( r - 1, \left\lceil \frac{M - r}{2} \right\rceil, \left\lfloor \frac{M - r}{2} \right\rfloor \right)$$

*of copies of  $(a, b, M)$ , whose budget is exactly  $(r - 1) + \lceil (M - r)/2 \rceil + \lfloor (M - r)/2 \rfloor = M - 1$ , has subset sums containing  $M$  consecutive integers.*

No restriction on  $r$  or  $a$  is imposed: a single closed-form multiset serves the whole line.

*Proof.* Any selection from the multiset is

$$v = ia + jb + kM = ia + wb + rak, \quad w := j - k$$

(using  $M = ra - b$ ), subject to  $i \leq x = r - 1$ ,  $k \geq \max(0, -w)$ ,  $k \leq z$  and  $w + k \leq y$ . So a  $(b, M)$  pair is a residue-neutral mover worth  $ra$ , and  $w$  is the number of *unpaired* copies of  $b$  (if  $w > 0$ ) or of  $M$  (if  $w < 0$ ).

Set

$$p := \left\lfloor \frac{a}{2} \right\rfloor, \quad q := \left\lfloor \frac{a - 1}{2} \right\rfloor, \quad \text{so} \quad p + q = a - 1,$$

and fix the run start  $\text{low} := \max(pb, qM)$ . We show that every  $n \in [\text{low}, \text{low} + M - 1]$  is realized.

*Preliminaries.* We record three consequences of the hypotheses, each valid for all  $a \geq 3$  and  $r \geq 3$ .

- (P1)  $M \geq a + r - 1$ . From  $b < M$  and  $b + M = ra$  we get  $ra \leq 2M - 1$ . Since  $a \geq 3$  and  $r \geq 3$  give  $a(r-2) \geq 3(r-2) \geq 2r-3$ , it follows that  $2M \geq ra+1 = a(r-2)+2a+1 \geq 2a+2r-2$ .
- (P2)  $M \leq a(r-1) - 1$ . From  $b > a$  we get  $M = ra - b < ra - a = a(r-1)$ .
- (P3)  $y \geq p$  and  $z \geq q$ . By (P1),  $M - r \geq a - 1$ , so  $z = \lfloor (M - r)/2 \rfloor \geq \lfloor (a - 1)/2 \rfloor = q$  and  $y = \lceil (M - r)/2 \rceil \geq \lceil (a - 1)/2 \rceil = \lfloor a/2 \rfloor = p$ .

*Residues, and where the four inequalities come from.* Since  $\gcd(a, b) = 1$ , for each target  $n \in [\text{low}, \text{low} + M - 1]$  there is a unique  $w \in [0, a - 1]$  with  $wb \equiv n \pmod{a}$ . We split on the size of  $w$ , and in each branch we exhibit the multiset explicitly and read off exactly what must be checked.

*Branch 1:  $w \leq p$  (spend  $w$  unpaired copies of  $b$ ).* Here  $wb \leq pb \leq \text{low} \leq n$ , so  $A := (n - wb)/a$  is a nonnegative integer. Write  $A = i + r\kappa$  with  $i := A \bmod r$ , so  $i \leq r - 1 = x$  and  $\kappa := \lfloor A/r \rfloor$ . Take the sub-multiset consisting of  $i$  copies of  $a$ ,  $w + \kappa$  copies of  $b$ , and  $\kappa$  copies of  $M$ . Its value is

$$ia + (w + \kappa)b + \kappa M = ia + wb + \kappa(b + M) = ia + wb + \kappa ra = a(i + r\kappa) + wb = aA + wb = n,$$

as required. It is available in the multiset  $(x, y, z)$  precisely when

$$\kappa \leq z \quad \text{and} \quad w + \kappa \leq y.$$

For the first:  $aA = n - wb \leq n \leq \text{low} + M - 1$ , and  $\kappa \leq z$  follows from  $A \leq rz + x$ , so it suffices that  $\text{low} + M - 1 \leq a(rz + x)$ . For the second: using  $arw = w(b + M)$ ,

$$a(A + rw) = (n - wb) + w(b + M) = n + wM \leq (\text{low} + M - 1) + pM,$$

and  $w + \kappa \leq y$  follows from  $A + rw \leq ry + x$ ; so it suffices that

$$\text{low} + M - 1 + pM \leq a(ry + x). \quad (\text{C}_y)$$

*Branch 2:  $w > p$  (spend  $v := a - w \in [1, q]$  unpaired copies of  $M$ ).* Since  $M \equiv -b \pmod{a}$ , we have  $vM \equiv -vb \equiv wb \equiv n \pmod{a}$ , and  $vM \leq qM \leq \text{low} \leq n$ , so  $A' := (n - vM)/a$  is a nonnegative integer. Writing  $A' = i' + r\kappa'$  as before, the sub-multiset of  $i'$  copies of  $a$ ,  $\kappa'$  copies of  $b$  and  $v + \kappa'$  copies of  $M$  has value  $n$ , and is available precisely when  $\kappa' \leq y$  and  $v + \kappa' \leq z$ . The first follows from  $\text{low} + M - 1 \leq a(ry + x)$ , which is weaker than  $(\text{C}_y)$ . For the second,  $a(A' + rv) = (n - vM) + v(b + M) = n + vb \leq (\text{low} + M - 1) + qb$ , so it suffices that

$$\text{low} + M - 1 + qb \leq a(rz + x). \quad (\text{C}_z)$$

(This also implies the condition  $\text{low} + M - 1 \leq a(rz + x)$  needed in Branch 1.)

*The four inequalities.* So the entire coverage claim reduces to the two conditions  $(\text{C}_y)$  and  $(\text{C}_z)$ . Each contains  $\text{low} = \max(pb, qM)$ , so each splits into two cases according to which term attains the maximum; using  $p + q = a - 1$  and  $b + M = ra$ , the four cases are exactly

$$\begin{aligned} (\text{E1}) \quad & (a - 1)b + M - 1 \leq a(rz + x) \quad [(\text{C}_z) \text{ with } \text{low} = pb], \\ (\text{E2}) \quad & q(b + M) + M - 1 \leq a(rz + x) \quad [(\text{C}_z) \text{ with } \text{low} = qM], \\ (\text{F1}) \quad & p(b + M) + M - 1 \leq a(ry + x) \quad [(\text{C}_y) \text{ with } \text{low} = pb], \\ (\text{F2}) \quad & aM - 1 \leq a(ry + x) \quad [(\text{C}_y) \text{ with } \text{low} = qM]. \end{aligned} \quad (19)$$

(For (E1):  $pb + qb + M - 1 = (a - 1)b + M - 1$ . For (F2):  $qM + pM + M - 1 = (a - 1)M + M - 1 = aM - 1$ .) Checking these four extremes suffices for *all*  $w$  and all  $n$  in the target interval, because the only place  $w$  entered was through the bounds  $w \leq p$  and  $v \leq q$ , which are exactly the extremes used. We now verify all four.

(F2). It suffices that  $M \leq ry + r - 1$ . From  $y \geq (M - r)/2$  and  $r \geq 3$ ,

$$ry + r - 1 \geq \frac{3}{2}(M - r) + r - 1 = \frac{1}{2}(3M - r - 2) \geq M \iff M \geq r + 2,$$

and  $M \geq a + r - 1 \geq r + 2$  by (P1) and  $a \geq 3$ .

(F1). Here  $p(b + M) = pra$ , and  $y \geq p$  by (P3) gives  $ary \geq pra$ . So (F1) reduces to  $M - 1 \leq a(r - 1)$ , which is (P2).

(E2). Identically,  $q(b + M) = qra$ , and  $z \geq q$  by (P3) gives  $arz \geq qra$ , so (E2) also reduces to (P2).

(E1). This is the binding inequality. Substituting  $b = ra - M$  and using  $z \geq (M - r - 1)/2$ , it suffices that

$$2[(a - 1)(ra - M) + M - 1] \leq ar(M - r - 1) + 2a(r - 1),$$

which, after expanding both sides, is

$$0 \leq M(ar + 2a - 4) - ar^2 - 2ra^2 + 3ar - 2a + 2.$$

The coefficient  $ar + 2a - 4$  is positive (as  $a, r \geq 3$ ), so we may insert the lower bound  $2M \geq ra + 1$  from (P1). Doing so and multiplying through by 2, the right-hand side is at least

$$(ra + 1)(ar + 2a - 4) - 2ar^2 - 4ra^2 + 6ar - 4a + 4 = a(ar^2 - 2ar - 2r^2 + 3r - 2).$$

Hence (E1) holds as soon as

$$ar(r - 2) \geq 2r^2 - 3r + 2. \quad (20)$$

For  $r \geq 4$ : since  $a \geq 3$  we have  $ar(r - 2) \geq 3r^2 - 6r$ , and  $3r^2 - 6r \geq 2r^2 - 3r + 2 \iff r^2 - 3r - 2 \geq 0$ , which holds for every  $r \geq 4$ . For  $r = 3$ , (20) reads  $3a \geq 11$ , i.e.  $a \geq 4$ .

So (20) can fail only for  $(a, r) = (3, 3)$ . But  $a = 3$  together with  $b + M = 3a = 9$  and  $3 < b < M$  forces  $b = 4$  and  $M = 5$ : the single triple  $(3, 4, 5)$ , for which  $(x, y, z) = (2, 1, 1)$  and (E1) reads  $2 \cdot 4 + 5 - 1 = 12 \leq 3(3 \cdot 1 + 2) = 15$ , true. (Concretely, the multiset  $\{3, 3, 4, 5\}$  has subset sums  $\{0, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15\} \supseteq [3, 9]$ , a run of  $7 \geq 5$  consecutive integers, at budget  $4 = M - 1$ .)

All four inequalities of (19) therefore hold for every hard-core triple on the line  $a \mid b + M$ , which completes the proof.  $\square$

*Remark 4.16.* Case P therefore requires no subset-sum search and no certificate table: a single closed-form multiset serves every triple on the line, and the only triple not covered by the estimate (20) is  $(3, 4, 5)$ , which is checked directly. In particular the three triples  $(3, 7, 8)$ ,  $(4, 9, 11)$ ,  $(5, 12, 13)$  — which earlier treatments carried as ad-hoc certificates for the  $r \geq 5$ ,  $M < 2a + 4$  corner — are ordinary instances of it, with  $x = r - 1 = 4$ .

The formalization takes a slightly different route here, for reasons of tactic convenience rather than mathematics: it discharges (19) for  $r \in \{3, 4\}$  (`caseP_r3`, `caseP_r4`), decides the finitely many  $r \geq 5$  triples with  $M < 2a + 4$  directly, and routes  $M \geq 2a + 4$  through the independent argument of Lemma 4.17. Either route closes the case; we record both.

**Lemma 4.17.** *If  $a \mid b + M$  and  $M \geq 2a + 4$ , then the hard-core triple  $(a, b, M)$  admits a multiset of budget  $\leq M - 1$  whose subset sums contain  $M$  consecutive integers.*

*Proof.* Since  $\bar{\mu} = a - \bar{e}$ , the representatives  $jb + kM \equiv \bar{e}(j - k) \pmod{a}$  cover  $\mathbb{Z}_a$  with any box  $(Y, Z)$  satisfying  $Y + Z = a - 1$ , as  $j - k$  then ranges over  $[-Z, Y]$ , a complete residue system. Choose  $Y = \lceil (a - 1)/2 \rceil$  and  $Z = \lfloor (a - 1)/2 \rfloor$ . The minimal representative of the class  $\bar{e}w$  has height  $wb$  (for  $w \geq 0$ ) or  $|w|M$  (for  $w < 0$ ), so  $S \leq \max(Yb, ZM)$ , and the budget satisfies  $\lceil (M - 1 + S)/a \rceil + (a - 1) \leq M - 1$  iff  $S \leq (a - 1)M - a^2 + 1$ .

The  $ZM$  branch gives  $ZM \leq \frac{a-1}{2}M \leq (a - 1)M - a^2 + 1$  once  $M \geq 2a + 2$ . For the  $Yb$  branch the parity of  $a$  enters. If  $a$  is odd then  $Y = Z = (a - 1)/2$  and the same computation applies. If  $a$  is even then  $Y = a/2$ , and the bound  $b \leq M - 1$  alone would require  $M \geq 2a + 3 + \frac{4}{a-2}$ , which exceeds  $2a + 4$  at  $a = 4$ . But  $b \leq M - 1$  is not tight on this line:  $a \mid b + M$  forces  $b + M \equiv 0$

(mod  $a$ ), so for even  $a$  the sum  $b + M$  is even, whence  $b \neq M - 1$  and therefore  $b \leq M - 2$ . With  $b \leq M - 2$  the  $Yb$  branch closes for  $M \geq 2a + 4$  as well. (In particular the  $a$ -even,  $b = M - 1$  corner is vacuous, not merely bounded.)  $\square$

**Proposition 4.18** (Exhaustion of Case P). *Every hard-core triple with  $a \mid b + M$  satisfies  $m(G) \leq M - 1$ .*

*Proof.* Write  $b + M = ra$ . Since  $a < b < M$  we have  $b + M > 2a$ , so  $r \geq 3$ , and Lemma 4.15 applies. (Note  $r = 2$ , i.e.  $M = 2a - b$ , is impossible, as is  $M = 2a$ , which would force  $a \mid b$ .)  $\square$

Lemma 4.15 already closes the case. We nonetheless record the following independent argument for the range  $M \geq 2a + 4$ , both because it is shorter there and because it is the route the formalization takes (Remark 4.16).

#### 4.6. Case L: $e = h = g$ .

**Lemma 4.19.** *For  $G = \{a, a + g, a + 2g\}$  with  $\gcd(a, g) = 1$  and  $g \leq a - 2$ , the multiset*

$$(x, y, z) = \left( \left\lfloor \frac{a-2}{2} \right\rfloor + 2g, 1, \left\lceil \frac{a-2}{2} \right\rceil \right),$$

*of budget  $a + 2g - 1 = M - 1$ , covers  $M$  consecutive integers.*

*Proof.* Here  $\mu = 2g$ ,  $(e', \mu') = (1, 2)$ ,  $C' = 0$ ,  $V' = 2z + 1$ ,  $c = z + 1$ . The offsets are  $je + k\mu = g(j + 2k)$ , and with  $y = 1$  we get  $\{j + 2k : j \leq 1, k \leq z\} = [0, 2z + 1]$ , contiguous. Note  $x - c = 2g - 2$  for odd  $a$  and  $2g - 1$  for even  $a$ . Three sub-cases.

*a even.* Here  $x - c = 2g - 1 \geq g$ , so  $x \geq c + g$ ; and with  $C' = 0$ ,  $V' = 2z + 1$ , the frame-merge condition of Lemma 4.4(d) reads  $V' - C' = 2z + 1 \geq a - 1$ , i.e.  $z \geq (a - 2)/2$  — exactly our  $z = (a - 2)/2$ , with equality. Both hypotheses of the extended form hold, so the sums cover the solid interval  $[(c + g - 1)a, (x - g + 1)a + g(2z + 1)]$ , of length  $a + g(a - 1) + 1 \geq M = a + 2g$ , since  $g(a - 1) \geq 2g - 1$  for  $a \geq 3$ .

*a odd,  $g \geq 2$ .* Here  $z = (a - 1)/2$  and  $x = c + 2(g - 1) \geq c + g$  (as  $g \geq 2$ ), and the frame-merge condition holds with room to spare:  $V' - C' = 2z + 1 = a \geq a - 1$ . The extended form gives the solid interval  $[(c + g - 1)a, (x - g + 1)a + g(2z + 1)]$ , of length  $g(2z + 1) + 1 = ga + 1 \geq a + 2g$ , since  $(g - 1)a \geq 2g - 1$  for  $g \geq 2$ ,  $a \geq 3$ .

*a odd,  $g = 1$ .* Here  $x = c$ , so Lemma 4.4(c) applies: level  $c$  covers  $ca + [0, V]$  with  $V = 2z + 1 = a$ , level  $c + 1$  covers  $(c + 1)a + [1, V]$ , and the merge condition  $V \geq a + \max(C, 1) - 1 = a$  holds with equality. The run  $[ca, (c + 1)a + V]$  has length  $2a + 1 \geq a + 2 = M$ , at budget  $2c = a + 1 = M - 1$ .  $\square$

#### 4.7. Case E: the universal $\eta$ -box.

**Lemma 4.20.** *Suppose  $a \nmid M$  and  $a \nmid b + M$  (so  $\eta \notin \{0, 1, a - 1\}$ ), and suppose  $a \geq 12$  and  $\mu = M - a \geq 12$ . Let  $t := \min(\eta, a - \eta) \in [2, a/2]$ ,  $Z := \lceil (a - t)/t \rceil$  and  $\sigma := t + Z$ . Then the box  $(Y, Z) = (t - 1, Z)$  covers  $\mathbb{Z}_a$  and its budget is  $\leq M - 1$ .*

*Proof. Coverage.* If  $t = \eta$ , the box offsets are  $\bar{e}(j + tk) \bmod a$  and  $\{j + tk : j \leq t - 1, k \leq Z\} = [0, t(Z + 1) - 1] \supseteq [0, a - 1]$ , since  $t(Z + 1) \geq a$ . If  $t = a - \eta$ , the offsets are  $\bar{e}(j - tk)$  and  $\{j - tk\} = [-tZ, t - 1]$ , an interval of length  $t(Z + 1) \geq a$ . Either way we obtain a complete residue system, multiplication by the unit  $\bar{e}$  being a bijection.

*Budget.* Write  $S_0 := (t - 1)b + ZM$  for the corner bound on the representative height, so  $S \leq S_0$ , and put  $K := t - 1 + Z = \sigma - 1$ . Then

$$B \leq \left\lceil \frac{M - 1 + S_0}{a} \right\rceil + K,$$

so it suffices that  $M-1+S_0 \leq a(M-1-K)$ : this forces  $\lceil (M-1+S_0)/a \rceil \leq M-1-K$  and hence  $B \leq M-1$ . Using  $b \leq M-1$  we have  $M-1+S_0 \leq M\sigma-t$ , while  $a(M-1-K) = a(M-\sigma)$ , so the sufficient inequality is in turn implied by  $M\sigma + a\sigma \leq aM + t$ , and hence by  $M(a-\sigma) \geq a\sigma$ .

*Threshold.* We have  $\sigma \leq t + (a-1)/t$ , whose maximum over  $t \in [2, a/2]$  is attained at an endpoint, by convexity:  $\sigma \leq \max(\frac{a+3}{2}, \frac{a}{2} + 2) = \frac{a}{2} + 2$ . Hence

$$\frac{a\sigma}{a-\sigma} \leq \frac{a(a+4)}{a-4} = a + 8 + \frac{32}{a-4} \leq a + 12 \quad (a \geq 12),$$

so  $M = a + \mu \geq a + 12$  suffices.  $\square$

**4.8. Case T:**  $\mu \leq 11$ . Fix  $(e, h)$  with  $\mu = e + h \leq 11$ ,  $e \neq h$  (else Case L),  $a \nmid M$  and  $a \nmid b + M$ . Since  $e \leq 10$  and  $h \leq 10$ , this case consists of finitely many *lines*, each parameterized by  $a$  alone. As  $e \neq h$  and  $e' < \mu'$ , here  $\mu' = \mu/g \geq 3$ .

**Lemma 4.21.** *With  $y = \mu' - 1$  (writing  $C := C'$  and  $V := V'$  in the  $g = 1$  variants, where  $e' = e$ ,  $\mu' = \mu$ , so  $C' = (e-1)(\mu-1)$ ) and*

- $g = 1$ , variant A:  $z = \max(e-1, \lceil (\max(a, \mu) - 1 + 2C - ye)/\mu \rceil)$  and  $x = c + 1$  (Lemma 4.4(b): merge  $V - C \geq a - 1$ , run length  $a + V - C + 1 \geq M$ );
- $g \geq 2$ :  $z = \max(e' - 1, \lceil (a + \lceil (\mu - 1)/g \rceil + 2C' - ye')/\mu' \rceil)$  and  $x = c + g$  (Lemma 4.4(d), base form — one frame per class, no merge hypothesis needed: solid run length  $g(V' - C') - (g-1)a + 1 \geq M$ );

*the construction covers  $M$  consecutive integers whenever the stated  $z$ -conditions hold, at budget  $2c + v$  with  $v \in \{1, g\}$ . Both variants are merge-robust: variant A uses the two-frame merge of Lemma 4.4(b), whose merge condition its choice of  $z$  enforces, and the  $g \geq 2$  variant uses the base form of Lemma 4.4(d), which needs no merge hypothesis. In particular Case T does not use the short merge, Lemma 4.4(c).*

*Proof.* Both variants instantiate Lemma 4.4 with the multiset  $(x, y, z)$  and  $y = \mu' - 1$ , whose budget is  $x + y + z$ ; we verify the hypotheses and the run length. Throughout,  $V' - C' = ye' + z\mu' - 2C'$ , and the two branches of the maximum defining  $z$  play distinct roles: the constant branch keeps  $z \geq e' - 1$ , the standing hypothesis of Lemma 4.4, while the ceiling branch supplies the run length.

*Variant A* ( $g = 1$ ; here  $e' = e$ ,  $\mu' = \mu$ ,  $C = C'$ ). The ceiling branch gives  $z\mu \geq \max(a, \mu) - 1 + 2C - ye$ , hence

$$V - C = ye + z\mu - 2C \geq \max(a, \mu) - 1 \geq a - 1,$$

which is exactly the two-frame merge condition of Lemma 4.4(b). With  $x = c + 1$ , that part covers  $[ca + C, (c+1)a + V]$ , an interval of length  $a + (V - C) + 1 \geq a + \max(a, \mu) \geq a + \mu = M$ ; so the subset sums contain  $M$  consecutive integers. The budget is  $x + y + z = 2c + 1$ .

*Variant  $g \geq 2$ .* The ceiling branch gives  $z\mu' \geq a + \lceil (\mu - 1)/g \rceil + 2C' - ye'$ , hence

$$V' - C' = ye' + z\mu' - 2C' \geq a + \left\lceil \frac{\mu-1}{g} \right\rceil \geq a + \frac{\mu-1}{g}.$$

With  $x = c + g$ , the base form of Lemma 4.4(d) — which needs *no* merge hypothesis — covers a solid interval of length

$$g(V' - C') - (g-1)a + 1 \geq g\left(a + \frac{\mu-1}{g}\right) - (g-1)a + 1 = a + \mu = M.$$

The budget is  $x + y + z = 2c + g$ .

Since  $g = \gcd(e, \mu) \geq 1$ , exactly one variant applies to each line, and both invoke only parts (b) and (d) of Lemma 4.4, never the short merge (c). The coverage hypotheses above hold for every



$a$ ; the only quantity that can violate the certificate is the budget  $2c + v$  ( $v \in \{1, g\}$ ), through its dependence on  $z$ .  $\square$

Only the budget inequality  $2c + v \leq M - 1$  can fail, and it fails only finitely often.

*Example 4.22* (a Case-T line). Take  $(e, h) = (2, 1)$ , so  $\mu = 3$ ,  $g = \gcd(2, 1) = 1$ ,  $(e', \mu') = (2, 3)$  and  $C' = (e' - 1)(\mu' - 1) = 2$ . This is a single line, parameterized by  $a$ : its members are the hard-core triples  $(a, a + 2, a + 3)$ . Variant A takes  $y = \mu' - 1 = 2$  and

$$z = \max\left(e - 1, \left\lceil \frac{\max(a, \mu) - 1 + 2C' - ye}{\mu} \right\rceil\right) = \max\left(1, \left\lceil \frac{a-1}{3} \right\rceil\right),$$

with  $c = y + z$  and  $x = c + 1$ , for a budget of  $2c + 1$  against the target  $M - 1 = a + 2$ . Selected values along the line (the table is a sample, not exhaustive):

| $a$ | $z$ | $c$ | budget $2c+1$ | target $a+2$ | margin    |
|-----|-----|-----|---------------|--------------|-----------|
| 7   | 2   | 4   | 9             | 9            | 0 (tight) |
| 13  | 4   | 6   | 13            | 15           | 2         |
| 19  | 6   | 8   | 17            | 21           | 4         |
| 25  | 8   | 10  | 21            | 27           | 6         |
| 31  | 10  | 12  | 25            | 33           | 8         |

The budget grows with slope  $2/\mu' = 2/3$  while the target grows with slope 1, so the margin increases at rate  $1/3$  and, once positive, stays positive. On this particular line the budget in fact never exceeds the target — it is merely tight at  $a = 7$  — so the line contributes *no* rows to Table A. Other lines are less lucky: the line  $(e, h) = (5, 1)$  contributes the Table-A rows  $(4, 9, 10)$ ,  $(7, 12, 13)$  and  $(8, 13, 14)$ , among others, where the budget does exceed  $M - 1$  and an explicit box certificate is needed. Lemma 4.23 is exactly the statement that, over all lines and all  $a \leq 3000$ , this happens for precisely 172 triples, and never for  $a > 3000$ .

**Lemma 4.23** (T-tail).

- (1) *For every line and every  $a \leq 3000$ , the minimal variant budget was computed exactly (the finite enumeration of Appendix A, whose failing triples are re-verified row-by-row by `gen-tables.py` and decided by the Lean kernel); it exceeds  $M - 1$  for precisely 172 triples, all with  $a \leq 29$  — these are Table A of Appendix B — and each of those carries a verified modulo- $a$  box certificate with  $Y + Z \leq a - 1$  and budget  $\leq M - 1$ .*
- (2) *For each line, the cheapest variant's budget is bounded above by the linear function*

$$\beta(a) := 2\left[(\mu' - 1) + \frac{a + \mu + 2C' - (\mu' - 1)e'}{\mu'} + 1\right] + g,$$

*of slope  $2/\mu' \leq 2/3$  in  $a$  (using  $\lceil u \rceil \leq u + 1$  and  $\mu' \geq 3$ ), while the target  $\tau(a) = a + \mu - 1$  has slope 1. (The  $z$ -formulas of Lemma 4.21 are maxima;  $\beta$  bounds their ceiling branch, and that branch is the active one throughout  $a \geq 3000$ : on every line its argument is at least  $2899/11 > 260$  — the numerator is at least  $3000 - 1 - ye \geq 2899$ , since  $y \leq \mu' - 1 \leq 10$  and  $e' \leq e \leq 10$  give  $ye \leq 100$ , and the denominator is  $\mu' \leq 11$  — whereas the competing branch,  $e - 1$  respectively  $e' - 1$ , is at most 9. Being increasing in  $a$  against a constant, the ceiling branch stays active for all  $a \geq 3000$ , so there the cheapest budget equals its ceiling value and  $\beta$  bounds it above.) The margin  $\tau - \beta$  is therefore nondecreasing in  $a$ , at rate at least  $1/3$ .*

*It remains to check that the margin is positive at  $a = 3000$ . We compare the target against  $\beta$ , the proved upper bound on the cheapest variant budget, and not against the exact budget; no separate reconciliation of exact budgets is therefore required. Evaluating*

$\beta(3000)$  on each of the finitely many lines and minimizing gives

$$\min_{\text{lines}} (\tau(3000) - \beta(3000)) = 995 > 0,$$

as reported by the verification harness of Appendix A. Since  $\tau - \beta$  is nondecreasing, it stays positive for every  $a > 3000$ . Hence there are no budget failures with  $a > 3000$ , and the failure set is exactly Table A.

**4.9. Case B: finite bases and the  $\lambda$ -lift.** For  $a \leq 11$  the classes  $(a, \bar{e}, h)$  with  $\bar{e} \in [1, a-1]$  and  $h \in [1, a-2]$  are finite in number, and the members of a class differ only in  $\lambda = \lfloor e/a \rfloor$ . For every class reaching this case, the member of smallest  $M$  is certified by an explicit modulo- $a$  box with  $Y + Z \leq a - 1$  and budget  $\leq M - 1$  — Table B of Appendix B, comprising 178 classes — and by Lemma 4.6 the same box certifies every larger member of the class.

Which branch a triple belongs to is class-invariant for D and P, since  $M \bmod a$  and  $(b + M) \bmod a$  depend only on  $(\bar{e}, h)$ ; and an  $e = h$  member can only be the  $\lambda = 0$  member. Table B lists bases accordingly, taking the first branch-B member of each class.

*Example 4.24* (a Case-B class and its lift). Take the class  $[a; \bar{e}, h] = [5; 1, 1]$ . Its base, listed in Table B, is  $(a, b, M) = (5, 16, 17)$  with box  $(x, Y, Z) = (10, 1, 2)$ : budget  $10 + 1 + 2 = 13 \leq M - 1 = 16$ . The members of the class are obtained by increasing  $\lambda = \lfloor e/a \rfloor$ , i.e. by  $(a, b, M) \mapsto (a, b + 5, M + 5)$ , giving  $(5, 21, 22)$ ,  $(5, 26, 27)$ , and so on — all with the same  $b \bmod 5$  and the same  $h = M - b = 1$ .

By Lemma 4.6 the *same* box  $(Y, Z) = (1, 2)$  certifies every one of them: the residues of  $jb + kM$  modulo 5 are unchanged, so the same representative pairs serve; each representative height grows by at most  $(Y + Z)a = 3 \cdot 5 = 15$ , so the budget grows by at most  $Y + Z + 1 = 4 \leq a = 5$ , while the target  $M - 1$  grows by exactly  $a = 5$ . The margin therefore never shrinks, and one finite check at the base settles the whole infinite class. This is why Table B lists 178 *classes* rather than infinitely many triples.

*Remark 4.25* (The descent). The claim that every larger member of a class is certified by the same box is a descent. A triple's class is  $(a, b \bmod a, h)$ , and the class-level branch conditions —  $M \bmod a$ ,  $(b + M) \bmod a$ , and the box's residue coverage — are invariant under the descent  $b \mapsto b - a$ ,  $M \mapsto M - a$  (this is where  $\gcd(a, b - a) = \gcd(a, b)$  and the modulo- $a$  recomputations enter, and  $h = M - b$  is unchanged). The condition  $e = h$  is *not* invariant:  $e = b - a$  drops by  $a$  at each step while  $h$  is fixed, so a member with  $e = h$  can only be the  $\lambda = 0$  (smallest) member of its class — which then belongs to Case L, its higher members ( $e > h$ ) belonging to Case B. This is precisely why the six diagonal classes below must be based at  $\lambda \geq 1$ . Iterating the descent lands on the listed base  $b_0 \leq b$  with  $b \equiv b_0 \pmod{a}$ , and Lemma 4.6 transports the box upward. Concretely, the enumeration for  $a \leq 11$  has 178 distinct classes, of which exactly six — the diagonal  $\bar{e} = h$  classes  $(9, 25, 32)$ ,  $(10, 27, 34)$ ,  $(11, 28, 34)$ ,  $(11, 29, 36)$ ,  $(11, 30, 38)$ ,  $(11, 31, 40)$  — must be based at a  $\lambda \geq 1$  member, because their  $\lambda = 0$  member has  $e = h$  and so belongs to Case L.

#### 4.10. Proof of Theorem 4.1.

*Proof.* We argue by strong induction on  $M$ : in proving the statement for a given  $M$  we may assume it for every alphabet of smaller maximum. This is what Lemma 4.13 consumes, passing to the rescaled alphabet  $H = G'/\delta$  of maximum  $M/\delta < M$ . No other branch appeals to the induction hypothesis, and no hard-core branch re-enters Lemma 4.13 at a larger  $M$ , so the recursion is well founded.

By Lemmas 4.7–4.13 it suffices to treat the hard core of §4.3. Given a hard-core triple  $(a, b, M)$ : if  $a \mid M$ , apply Lemma 4.14; else if  $a \mid b + M$ , apply Lemma 4.15 (Proposition 4.18); else if

$e = h$ , apply Lemma 4.19; else if  $a \geq 12$  and  $\mu \geq 12$ , apply Lemma 4.20; else if  $\mu \leq 11$ , apply Lemmas 4.21 and 4.23 together with Table A; else  $a \leq 11$  and  $\mu \geq 12$ , and we apply Table B with Lemma 4.6.

*Exhaustiveness.* Cases D and P are disjoint lines (§4.3). A triple with  $e = h$  and  $a \mid b + M$  would need  $a \mid 3e$ , possible only for  $a = 3$ ,  $e = 1$ , i.e.  $(3, 4, 5)$ , where both Lemma 4.15 (witness  $(2, 1, 1)$ ) and Lemma 4.19 apply. Every remaining triple has  $\eta \notin \{0, 1, a-1\}$  and falls into exactly one of E, T, B according to the sizes of  $a$  and  $\mu$ : either both are  $\geq 12$  (Case E), or  $\mu \leq 11$  (Case T), or  $\mu \geq 12$  and  $a \leq 11$  (Case B). Each branch delivers a multiset of size  $\leq M - 1$  whose subset sums contain  $M$  consecutive integers.  $\square$

## 5. ASSEMBLY: PROOF OF THEOREM A

*Proof of Theorem A.* Part (1) is Theorem 2.1.

For part (2), let  $d_2 \leq k$  with  $k \geq 3$ , and let  $(r_i)$  be any prescribed growth sequence. Build the certificate  $B$  of Lemma 3.1. Let  $A$  be any  $(d_1, d_2)$ -sequence, let  $G_\infty$  be its tail alphabet and  $g := \gcd(G_\infty)$ . By Lemma 3.2 we may pass to the tail, and by Lemma 3.4 we may assume  $\gcd(G_\infty) = 1$  after rescaling. Then:

| Case                                                                           | Handled by                                                                                                         |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $ G_\infty/g  = 1$                                                             | Lemma 3.3 (AP tail)                                                                                                |
| $ G_\infty/g  = 2$ , gap word eventually periodic                              | Lemma 3.5                                                                                                          |
| $ G_\infty/g  = 2$ , some $W_2(\sigma_0) \geq 2$ , $k$ odd or $d_2 \leq k - 1$ | Lemmas 3.8(a) and 3.7                                                                                              |
| $ G_\infty/g  = 2$ , some $W_2(\sigma_0) \geq 2$ , $k$ even, $d_2 = k$         | Lemma 3.8(a) boundary routing                                                                                      |
| $ G_\infty/g  = 2$ , all $W_2 \leq 1$ (balanced)                               | Lemma 3.8(b): Prop. 3.9, then<br>Lemma 3.5 or 3.11                                                                 |
| $ G_\infty/g  \geq 3$                                                          | Lemma 3.14 (slot lemma), with<br>$m(G_\infty/g) \leq \max(G_\infty/g) - 1 \leq d_2 - 1 \leq k - 1$<br>by Theorem B |

In every case  $kA$  is tail-covering (modulo  $g \cdot m'$ , after Lemma 3.4) — this is Proposition 3.16 — so  $B$  meets  $kA$  by Lemma 3.1. Since the growth sequence  $(r_i)$  was arbitrary, non-existence holds in the strong form asserted.

Combining (1) and (2): if  $d_2 \geq k + 1$  then  $r_k(d_1, d_2)$  exists, and if  $d_2 \leq k$  it does not. As exactly one of these holds for any  $k, d_2$ , the dichotomy follows.  $\square$

## 6. FORMAL VERIFICATION IN LEAN 4

The entire dichotomy has been formalized in Lean 4 against Mathlib, and machine-checked. The development, the verification scripts and build instructions are available at [https://github.com/beetree/math\\_erdos\\_1112](https://github.com/beetree/math_erdos_1112). This section records the formal statement, argues its faithfulness to the informal problem, and states the trust base. Appendix C maps each load-bearing result above to its Lean declaration.

A *sorry-free* proof of the *wrong* statement proves nothing, so the encoding is given in full and justified. The definitions live in a frozen statement file `Erdos1112.lean`, which is imported by the development but never edited by it; the three theorems are stated once each, directly over those definitions, in `Erdos1112Proof/Final.lean`. There is a single copy of each definition, so what is audited below is literally what the theorems assert.

**6.1. The encoding.** Sequences  $A, B$  are functions  $\mathbb{N} \rightarrow \mathbb{N}$ , 0-indexed (so  $b_0$  is  $b_1$ ).

```
def IsLacunaryWith (r : ℕ) (b : ℕ → ℕ) : Prop :=
  0 < b 0 ∧ StrictMono b ∧ ∀ i, r * b i ≤ b (i + 1)
```

```

def HasGapsIn (d1 d2 : ℕ) (a : ℕ → ℕ) : Prop :=
  0 < a 0 ∧ ∀ i, a i + d1 ≤ a (i + 1) ∧ a (i + 1) ≤ a i + d2

def kFoldSumset (k : ℕ) (a : ℕ → ℕ) : Set ℕ :=
  { n | ∃ f : Fin k → ℕ, n = ∑ j, a (f j) }

def RatioWorks (k d1 d2 r : ℕ) : Prop :=
  ∀ b : ℕ → ℕ, IsLacunaryWith r b →
  ∃ a : ℕ → ℕ, HasGapsIn d1 d2 a ∧
  Disjoint (kFoldSumset k a) (Set.range b)

def Question (k d1 d2 : ℕ) : Prop := ∃ r : ℕ, RatioWorks k d1 d2 r

def IsVarLacunaryWith (R : ℕ → ℕ) (b : ℕ → ℕ) : Prop :=
  0 < b 0 ∧ StrictMono b ∧ ∀ i, R i * b i ≤ b (i + 1)

```

These are reproduced verbatim from the source. The three theorems are then, in `Final.lean`:

```

theorem erdos_1112 (k d1 d2 : ℕ) (hk : 3 ≤ k) (hd1 : 1 ≤ d1) (hd : d1 < d2) :
  Question k d1 d2 ↔ k + 1 ≤ d2

theorem erdos_1112_existence_bound (k d1 d2 : ℕ) (hk : 3 ≤ k) (hd1 : 1 ≤ d1)
  (hd : d1 < d2) (h : k + 1 ≤ d2) : RatioWorks k d1 d2 (192 * d2)

theorem erdos_1112_strong_nonexistence (k d1 d2 : ℕ) (hk : 3 ≤ k)
  (hd1 : 1 ≤ d1) (hd : d1 < d2) (h : d2 ≤ k) (R : ℕ → ℕ) :
  ∃ b : ℕ → ℕ, IsVarLacunaryWith R b ∧
  ∀ a : ℕ → ℕ, HasGapsIn d1 d2 a →
  (kFoldSumset k a ∩ Set.range b).Nonempty

```

**6.2. Why the encoding is faithful.** A handful of modeling choices carry the weight.

- (1) **Infinite increasing sets as strictly increasing functions.**  $A$  and  $B$  are infinite sets of positive integers, encoded by their increasing enumerations. `StrictMono` is what makes the function enumerate a *set* rather than a multiset; it is imposed on  $B$  directly and *derived* for  $A$  from the gap condition when  $1 \leq d_1$  (the lemma `HasGapsIn.strictMono`).
- (2) **The gap window, phrased additively.** Over  $\mathbb{N}$ , subtraction is truncated. The definition writes  $d_1 \leq a_{i+1} - a_i \leq d_2$  as  $a_i + d_1 \leq a_{i+1} \wedge a_{i+1} \leq a_i + d_2$ . This avoids a real trap: with a subtractive phrasing, on a *decreasing* step  $a_{i+1} - a_i$  underflows to 0, so an isolated upper bound  $a_{i+1} - a_i \leq d_2$  would read  $0 \leq d_2$  — true — silently admitting negative gaps. (The lower bound is not the trap: an underflowed gap gives  $d_1 \leq 0$ , false, since  $1 \leq d_1$ .)
- (3) **The sumset, with repetition.** `kFoldSumset` ranges over an arbitrary index function  $f : \text{Fin } k \rightarrow \mathbb{N}$  with *no* injectivity requirement, so summands may repeat. Requiring  $f$  injective would model the distinct-summand sumset, a strictly smaller set, and a different problem.
- (4) **Quantifier order.** `Question` is  $\exists r \forall B \exists A$ , with  $A$  chosen after  $B$  — not the easier  $\exists A \forall B$ .
- (5) **The ratio over  $\mathbb{N}$  rather than  $\mathbb{Z}$ .** The problem says “an integer  $r$ ”. Passing to  $\mathbb{N}$  loses no existence content: a natural witness is an integer witness; and given an integer witness  $r < 0$ , the ratio condition is vacuous (as  $B$  is positive,  $rb_i \leq 0 < b_{i+1}$ ), so the admissible class of  $B$  coincides with that at  $r = 0$ , and 0 is a natural witness. The lemma `RatioWorks.mono` — a larger  $r$  only shrinks the admissible class of  $B$  — is one machine-checked ingredient. The full  $\mathbb{Z} \rightarrow \mathbb{N}$  bridge is now also formalized: the frozen

file carries the integer form `QuestionInt` and the theorem `question_iff_questionInt`, that `Question`  $\leftrightarrow$  `QuestionInt`, so the reduction is a proved equivalence, not only an informal justification (§6.4).

- (6) **A genuine, non-vacuous biconditional.** Both directions are separately witnessed, and both sides of the iff are inhabited within the allowed parameter family: when  $d_1 \leq k$ , the boundary  $d_2 = k + 1$  realizes the existence side and any  $d_1 < d_2 \leq k$  realizes the non-existence side. None of the predicates is silently trivial: `IsLacunaryWith`  $r$  is satisfiable for every  $r$  (take  $b_i = (r + 2)^i$ ), `HasGapsIn` is satisfiable (take  $a_i = 1 + d_1 i$ ), and `kFoldSumset`  $k$   $a$  is nonempty while `Set.range`  $b$  is infinite, so the disjointness requirement is a genuine constraint.
- (7) **Strong non-existence.** `erdos_1112_strong_nonexistence` says strictly more than  $\neg$ `Question`: for *every* ratio sequence  $R$  there is a *single*  $B$ , lacunary with those varying ratios, colliding with the sumset of every admissible  $A$ . Specializing  $R \equiv r$  recovers  $\neg$ `RatioWorks`  $k$   $d_1$   $d_2$   $r$  for every  $r$ , hence  $\neg$ `Question`; the bridge is the lemma `isVarLacunaryWith_const_iff`.

**6.3. Trust base.** We state the claim in the form in which it is checkable. *At the repository commit recorded in §9, built with the pinned toolchain `leanprover/lean4:v4.27.0` and the Mathlib revision fixed by `lake-manifest.json`, the command `#print axioms` reports, for each of the three headline theorems, exactly the list below. The development is sorry-free: a source scan finds no `sorry`, no `axiom` declaration, and no `native_decide` anywhere. Accordingly the audit reports*

[propext, Classical.choice, Quot.sound]

— the three standard foundational axioms of Lean and Mathlib, and nothing else. In particular there is no `sorryAx` (no gaps) and no `Lean.ofReduceBool`, the latter being the axiom that `native_decide` would introduce. Every finite and decidable check — the certificate tables of Proposition A.1 and the  $a \leq 3000$  Case-T scan — is therefore closed by the trusted *kernel*, not by compiled code: the trusted base is Lean plus Mathlib, and not the Lean compiler. The toolchain is pinned (`leanprover/lean4:v4.27.0`, with Mathlib pinned via `lake-manifest.json`), so the audit is reproducible from a clean checkout:

```
lake build                                # compiles the development
grep -rEn '\bsorry\b' --include='*.lean' Erdos1112Proof \
  | grep -v -- '--' | grep -v 'sorry-free'    # expect no output
#print axioms Erdos1112.erdos_1112          # expect the 3-axiom list
```

Two heavyweight declarations set a raised kernel heartbeat locally (`tailCovering_of_sturmian` and `exists_safe_subinterval`); this is an elaboration budget only, and carries no trust cost.

A caveat on what this does and does not say. “Depends only on three axioms” is a statement about the *logical* axiom report. The operative trusted base is larger: it includes the Lean kernel and elaborator, the Mathlib source at the pinned revision, and the dependency snapshot. And an axiom audit cannot detect a mismatch between a numbered lemma in this paper and the declaration that purports to formalize it; that correspondence is asserted in Appendix C and must be checked by reading both. Where the formal route differs from the prose route we say so explicitly (Remark 4.16 and §6.5).

**6.4. Scope of the formal development.** We record two points about scope, because the reader is entitled to know exactly what the machine checks — and, in one case, what it deliberately does not.

- (1) **The  $\mathbb{Z} \rightarrow \mathbb{N}$  bridge.** The problem asks for an *integer*  $r$ ; the development quantifies  $r$  over  $\mathbb{N}$ . The frozen statement file now also carries the integer form,

```

def IsLacunaryWithInt (r : ℤ) (b : ℕ → ℕ) : Prop :=
  0 < b 0 ∧ StrictMono b ∧ ∀ i, r * (b i : ℤ) ≤ (b (i + 1) : ℤ)

def QuestionInt (k d1 d2 : ℕ) : Prop := ∃ r : ℤ, RatioWorksInt k d1 d2 r

```

together with the machine-checked equivalence `question_iff_questionInt`, that `Question`  $\leftrightarrow$  `QuestionInt`, and the headline theorem in integer form,

```

erdos_1112_int : QuestionInt k d1 d2  $\leftrightarrow$  k + 1 ≤ d2.

```

So the dichotomy is verified for the problem’s literal integer phrasing, not merely for a natural-number rendering of it.

- (2) **The case  $k = 2$  of Theorem 2.1.** Theorem 2.1 is proved here in prose for all  $k \geq 2$ , but the formalization covers  $k \geq 3$  only — `existence_bound` and `erdos_1112_existence_bound` carry the hypothesis  $3 \leq k$ . This is deliberate rather than a gap to be closed: the problem fixes  $k \geq 3$ , so the formal development carries exactly the problem’s hypotheses throughout, and the  $k = 2$  existence statement stands as an unformalized supplementary observation (Remark 2.3).

The verification claim is therefore: *Theorem A for  $k \geq 3$ , in both the natural and the integer phrasing, is machine-checked over the definitions of §6 at the release named in §9.* Every headline Lean theorem carries the problem’s own hypothesis  $3 \leq k$ . Nothing in the mathematical argument of this paper is delegated to the formalization.

**6.5. Deviations of the formalization from this text.** The current paper and formalization agree on every mathematical statement. The deviations below are differences of *route* — the formalization sometimes proves the same conclusion a different way — not disagreements about what is true. (During development, formalization did expose a missing hypothesis in an earlier draft of Lemma 4.4(d); the present statement includes it, and its necessity is witnessed by the counterexample in Remark 4.5.) They have all been incorporated into the text above.

- (1) Case P is proved by one  $r$ -parameterized construction (Lemma 4.15), replacing separate  $r = 3$  and  $r = 4$  families, three ad-hoc  $r = 5$  certificates, and a finite check for  $a < 15$ . The reach inequalities (19) close algebraically for every  $a \geq 3$  and  $r \geq 3$  satisfying (20), and the only triple escaping that estimate is  $(3, 4, 5)$ , checked inline. No subset-sum search occurs in Case P, in the text or in the formalization, and no certificate table is needed. In Lean the construction is `caseP_pair` (uniform in  $r$ , taking (19) as hypotheses), instantiated by `caseP_r3` / `caseP_r4` and completed by `caseP_large`; see Remark 4.16.
- (2) Case D closes uniformly on  $(a - 1)(q - 2) \geq 0$ , with no split on  $q$  (Lemma 4.14).
- (3) The two-letter core takes a density-free route: the balanced classification via a discrepancy-oscillation iteration (Proposition 3.9), and the uniform syndeticity of Step 1 of Lemma 3.11 via an explicit Dirichlet-step circle walk, rather than the three-distance theorem. Dirichlet’s approximation theorem is the only analytic input.
- (4) The free-gap lemma is proved at the single index  $s_1 = \lceil t/\text{hi} \rceil$  (where  $\text{hi}$  is the upper endpoint of the parent interval; cf. Lemma 2.2), with safety for all  $s$  by monotonicity (Lemma 2.2).
- (5) Case T is run through the two merge-robust staircase variants only, in the text and in the formalization alike; neither uses Lemma 4.4(c). Table A is the failure set of that route, and is the only certificate table Case T needs.
- (6) The strong non-existence theorem is stated in the constructive `Set.Nonempty` form.



Formalizing the tables and scans exposed no erroneous, missing or redundant object. The extended form of Lemma 4.4(d), whose pre-repair statement was false, is carried in its repaired form, and its counterexample (5, 7, 9) (Remark 4.5) is pinned as a decided guard.

## 7. PROVENANCE AND AUTHORSHIP

This work is human-orchestrated but was not carried out unaided, and we state the division of labour plainly, since a reader is entitled to know how the argument was produced before deciding how to check it.

The core mathematics — the constructions, the case analyses, and the proofs of the lemmas above, together with the Lean formalization and the drafting of this paper — was carried out by Claude (Anthropic; models Fable 5 and Opus 4.8) acting as the reasoning agent, under the author’s direction. GPT-5.5 (OpenAI) and Gemini 3.1 (Google DeepMind) were consulted for advice and for adversarial review of drafts.

The author’s role was that of principal investigator, not scribe. He selected the problem and fixed its formalization (the Erdős Problems phrasing of §1); set the intermediate targets and the overall strategy; decided which candidate routes to pursue and which to abandon; ran the cross-model review cycles; audited each construction and proof; chose the formal statement whose fidelity is defended in §6; and made the final mathematical judgments. He takes full responsibility for the content, including any error that remains. We are deliberately exact about one point: the correctness of the argument does *not* rest on a claim that a human re-derived every lemma unaided. It rests on the Lean kernel (§6), which checks the proof independently of who or what produced it, and on the finite certificates, which anyone can re-verify. The automated systems are tools, not authors — they meet no authorship criterion and cannot take responsibility for the work — and they are not listed as authors.

An argument produced this way warrants more scrutiny, not less. That is the reason the development has been formally verified in Lean 4 (§6) rather than merely asserted: the kernel is indifferent to the provenance of a proof, and a reader who distrusts the origin of this argument is not asked to extend trust, but invited to run the check. Its exact scope is stated once, in §6.4.

That said, formal verification settles only correctness relative to the formal statement. It does not establish novelty, attribution, or exposition, and it does not substitute for the author’s understanding of the argument. The result may be checked by three *complementary* routes — reading the prose of §§2–5 together with the certificate tables of Appendix B; running the Python harnesses of Appendix A on the finite layer; or building the Lean development and inspecting the axiom audit (§6.3). We call these complementary rather than independent: they share the same mathematical architecture, the same case split, and the same witness constructions, so they give defence in depth against different failure modes, not three statistically independent confirmations. The one bridge no formalization can discharge on the reader’s behalf — that the formal statement means what the informal problem says — is set out decision by decision in §6, and is where we invite the sharpest scrutiny.

## 8. DECLARATIONS

**Funding.** None.

**Competing interests.** The author declares no competing interests.

**Author contributions.** Sole author. The division of labour between the author and the automated systems used is set out in §7.

**Use of generative AI.** The core mathematical constructions and proofs were produced by Claude (Anthropic; models Fable 5 and Opus 4.8) acting as a reasoning agent, under the author’s direction. GPT-5.5 (OpenAI) and Gemini 3.1 (Google DeepMind) were consulted for

advice and adversarial review. The author set the problem and the intermediate targets, audited the output at every stage, and is responsible for the content, including any error that remains. The automated systems are not authors: they cannot take responsibility for the work, and they meet no authorship criterion. Every mathematical claim in this paper has been checked by at least one of the three routes of §7, and the dichotomy of Theorem A is machine-checked in Lean 4 within the scope stated in §6.4. This declaration should be adapted to the requirements of the venue before submission; policies differ, and some require the exact model versions, the tasks for which each was used, and a statement of how the output was verified — all of which are given here and in §7.

*A note on repository metadata.* The version-control history of the accompanying repository contains Co-Authored-By trailers naming the automated system. These record which tooling produced a commit; they are not a claim of scholarly authorship, which is as stated above.

**Data and code availability.** See §9.

## 9. DATA AVAILABILITY AND REPRODUCTION

Everything needed to check this paper is in the repository [https://github.com/beetree/math\\_erdos\\_1112](https://github.com/beetree/math_erdos_1112). The stable `main` branch carries the Lean 4 development [Lean4, Mathlib], at commit `73af91479694cb8eddf5b1521d80f8f7aba8d2a6`; the complete package for the present paper — this L<sup>A</sup>T<sub>E</sub>X source (article and supplement), the machine-readable certificate exports (`table-A.csv`, `table-B.csv`, `SHA256SUMS`), the table generator/re-verifier `gen-tables.py`, and the two Python harnesses — is on the branch `paper-review`.

*The machine-checked proof.* A clean checkout builds with the pinned toolchain (`leanprover/lean4:v4.27.0`; dependency revisions fixed in `lake-manifest.json`): `lake exe cache get` then `lake build` compiles every file and prints the axiom audit of §6.3, reporting exactly [`propext`, `Classical.choice`, `Quot.sound`] for the headline theorems — including `erdos_1112_int`, the literal integer form.

*The finite layer.* The complete certificate tables are typeset in the supplementary document `erdos1112-supplement.pdf` and exported as `data/table-A.csv` and `data/table-B.csv` (digests in `data/SHA256SUMS`); the article itself gives the definition and a worked example of each (appendix on the certificate tables). Each row is a self-certifying subset-sum witness, mechanically verifiable from its tuple, and the Lean kernel re-decides all of them in the build above. Running `make` in `paper/` regenerates both the supplement and the exports from the one canonical source and re-checks every row (budget  $\leq M - 1$  and coverage of  $M$  consecutive integers, by exact bitmask, aborting on any failure), so no listing — printed, exported, or kernel-decided — can silently diverge. The two Python harnesses of Appendix A reproduce the branch counts and scans of Proposition A.1.

**Submission artifact.** This version is archived immutably in the repository [https://github.com/beetree/math\\_erdos\\_1112](https://github.com/beetree/math_erdos_1112) as the annotated Git tag `submission-2026-07`, which resolves to a single commit carrying this L<sup>A</sup>T<sub>E</sub>X source, the machine-readable certificate data, the generator/re-verifier, and the two Python harnesses; the Lean 4 development is the `main`-branch commit cited above. This PDF is built from the tagged commit by a clean checkout. We take the commit hash to which `submission-2026-07` resolves (recoverable by `git rev-parse`) as the permanent identifier of record, in preference to a DOI: a Git commit fixes the source, the formal proof, and the certificate data together, and is verifiable by anyone with the repository. The full certificate tables are typeset in the supplement `erdos1112-supplement.pdf` and exported to `data/`.

## APPENDIX A. THE FINITE LAYER

We isolate the entire finite content of the proof as a single statement, so that it is auditable at once rather than scattered through the argument.

**Proposition A.1** (Finite verification). *The finite certificate set — Table A, Table B and the Table-B supplement — has the following properties, each a finite, mechanical check:*

- (1) *every certificate has budget  $\leq M - 1$ ;*
- (2) *the subset sums of every certificate contain  $M$  consecutive integers;*
- (3) *the Case-T line scan ( $\mu \leq 11$ ) over  $a \leq 3000$  produces exactly the 172 exceptional triples of Table A, and none for  $a > 3000$ ;*
- (4) *the Case-B base classes ( $a \leq 11$ ) are complete;*
- (5) *the  $\lambda$ -lift (Lemma 4.6) applies to every larger member of each Table-B class.*

*Proof.* (i)–(ii) are direct finite checks: for each row, sum the budget and compute its subset-sum set by exact arbitrary-precision bitmask, verifying that it contains a run of length  $M$ . This is where Lemma 4.3 enters: a modulo- $a$  box with  $Y + Z \leq a - 1$  certifies exactly such a run. (iii)–(iv) are exhaustive bounded scans — the  $a \leq 3000$  line scan of Lemma 4.23(i) and the  $a \leq 11$ , bounded- $b$  class enumeration of §4.9 — made complete by the slope-tail argument of Lemma 4.23(ii) for  $a > 3000$  and by the descent of Remark 4.25 for larger  $b$ . Finally (v) is Lemma 4.6. Note that Case P contributes nothing to this proposition: by Remark 4.16 it is closed in the text, with no certificate.  $\square$

**A.1. What is checked, and how.** Three kinds of computation participate, and no others.

*Finite certificate tables* (Appendix B): 172 + 178 explicit instances of the proved frame lemma (Lemma 4.3). Each is checked by exact subset-sum computation with arbitrary-precision bitmasks, and Lemma 4.6 extends each Table-B certificate to the infinitely many larger members of its class. Case P contributes *no* certificates: it is closed in prose (Lemma 4.15 and Remark 4.16). Earlier versions of this work carried three ad-hoc Case-P witnesses; these are now ordinary instances of the uniform construction and no longer appear as data.

*Finite scans closing crude constants of proved lemmas:* the Case-T line scan to  $a = 3000$  with the slope-tail argument (Lemma 4.23); and the minimal-alphabet enumeration to  $M \leq 130$  (Lemmas 4.12 and 4.13). Case P requires no scan (Remark 4.16).

*End-to-end redundant verification* (not logically required, but performed): every one of the 83,251 hard-core triples with  $M \leq 120$  was covered by its designated branch with an exactly verified multiset — zero failures — by two separate implementations. The first, `sharp6_final.py`, is the constructive harness. The second, `sharp_referee_check.py`, is a re-implementation that shares no code with the first and reconstructs each designated witness from the lemma statements alone, with no auxiliary searches; it reports identical branch counts, and additionally re-verifies Lemmas 4.2–4.6 and 4.8–4.11 by brute force on thousands of instances, re-validates both tables including base-minimality and class-completeness, walks the  $\lambda$ -lift chains over 29,402 members arithmetically to  $M \leq 1500$  and 4,574 exactly to  $M \leq 250$ , checks the pair families for all  $a \leq 120$ , and checks the  $\eta$ -box algebra to  $a \leq 400$  at the worst member  $M = a + 12$  (which, being monotone in  $M$ , confirms Lemma 4.20 for all  $M$ ). All with zero failures. A further 800 random hard-core triples with  $M \in (120, 200]$  were spot-checked against the designated branches, again with zero failures.

*Scope of the independence.* We should not overstate what the second harness buys. It is best described as an *independent implementation of the same certificate specification*: it shares no code with the first and rebuilds each witness from the lemma statements, but it is *not* independent of the mathematical case split, the witness design, or the branch structure — all of which are the ones stated in §4. Agreement between the two therefore rules out coding slips,

not errors in the case analysis itself. It corroborates the finite layer only, and says nothing about the asymptotic lemmas, which are established in the text and formalized in Lean.

**A.2. Kernel re-checking of the finite layer.** The formalization re-verifies the entire finite layer inside the Lean kernel, with no `native_decide` and no external process, so the Python harnesses are corroborated rather than relied upon.

The paper and the formalization now follow *one* route through Case T (the merge-robust variants; see Lemma 4.21), so Table A is the same object in both, with the same 172 rows. The only count that differs is a transcription artifact in Table B, and we record it.

| Dataset          | Printed | Kernel rows | Reason for any difference                                                                                                                                                                                                        |
|------------------|---------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Table A (Case T) | 172     | 172         | none: identical                                                                                                                                                                                                                  |
| Table B (Case B) | 178     | 202         | 172 main rows + a 30-row supplement. Of the supplement, 24 duplicate main rows and 6 are new bases (the six diagonal $\bar{e} = h$ classes of Remark 4.25); so $172 + 6 = 178$ <i>distinct</i> classes, which is what is printed |
| Case P           | 0       | 0           | closed in prose; no certificate needed                                                                                                                                                                                           |
| Total            | 350     | 374         | the 24 duplicated Table-B rows                                                                                                                                                                                                   |

So the whole difference is the 24 rows that the Lean transcription of Table B carries twice. There is no longer any route divergence in the certificate data: the prose proof and the formal proof use the same lemmas, the same variants and the same tables. All of this data is exported in machine-readable form (§9).

The  $a \leq 3000$  Case-T scan is likewise redone as kernel `decide` sweeps, in chunks of 1000 values.

Finally, the certificate tables (typeset in full in the supplement, exported to `data/`, and exemplified in Appendix B) are generated mechanically from the same source data that the Lean development transcribes, and every row is re-verified at generation time (budget and coverage, by exact bitmask), so no listing can silently diverge from the checked one.

## APPENDIX B. THE CERTIFICATE TABLES

Each entry is a hard-core triple  $(a, b, M)$  together with a *box certificate*  $(x, Y, Z)$ , asserting that the multiset  $S$  consisting of  $x$  copies of  $a$ ,  $Y$  copies of  $b$  and  $Z$  copies of  $M$  has

- **budget**  $x + Y + Z \leq M - 1$ , and
- **coverage**:  $\text{subset-sums}(S)$  contains  $M$  consecutive integers,

i.e. that  $S$  witnesses  $m(G) \leq M - 1$  for  $G = \{a, b, M\}$ . The mechanism is Lemma 4.3: the copies of  $b$  and  $M$  realize a modulo- $a$  box  $(Y, Z)$  with  $Y + Z \leq a - 1$  covering  $\mathbb{Z}_a$ , and the  $x$  copies of  $a$  lift that residue cover to a solid run  $[S, S + M - 1]$ .

*Example B.1.* Row  $(5, 8, 9) \rightarrow (5, 2, 1)$ : the multiset  $\{5, 5, 5, 5, 5, 8, 8, 9\}$  has budget  $5 + 2 + 1 = 8 = M - 1$ . Its subset sums  $\{5i + 8j + 9k : i \leq 5, j \leq 2, k \leq 1\}$  hit every residue modulo 5 — the parts  $8j + 9k \in \{0, 8, 16, 9, 17\}$  have residues  $\{0, 3, 1, 4, 2\}$  — and, lifted by the copies of 5, contain the run  $[17, 25]$ , which is  $9 = M$  consecutive integers. That single computation is exactly what the checker performs on each row.

Table A lists the 172 exceptional lines of Case T (Lemma 4.23), all with  $a \leq 29$ . Table B lists one base per class  $(a, \bar{e}, h)$  for Case B, extended to all larger members by Lemma 4.6 (the Lean transcription additionally re-bases the six diagonal classes of Remark 4.25). A representative Table-B row is the class  $[5; 1, 1]$  with base  $(a, b, M) = (5, 16, 17)$  and box  $(x, Y, Z) = (10, 1, 2)$ :

budget  $10 + 1 + 2 = 13 \leq M - 1 = 16$ , and the parts  $16j + 17k$  ( $j \leq 1, k \leq 2$ ) realize every residue modulo 5, which the copies of 5 lift to a run of  $17 = M$  consecutive integers (cf. Example 4.24).

Rather than fill several pages of the article with  $172 + 178$  dense rows — each an independently machine-checkable tuple, of little value read by eye — we provide the *complete* tables as supplementary material: the typeset listing `erdos1112-supplement.pdf` (built by the same `make`), and the machine-readable exports `data/table-A.csv` and `data/table-B.csv`, whose SHA-256 digests are recorded in `data/SHA256SUMS`. All are generated from the one canonical source and re-verified row by row by `gen-tables.py` (§9), and the same rows are decided by the Lean kernel (§6.3); the tables are thus present in full, simply not inlined here.

## APPENDIX C. CORRESPONDENCE WITH THE LEAN DEVELOPMENT

Declarations live in the namespace `Erdos1112.Proof`, in files relative to `Erdos1112Proof/`, except the three headline theorems, which are in namespace `Erdos1112`. Every listed result is load-bearing: the development formalizes the full case tree rather than citing anything externally.

The mapping is *not* a claim of lemma-by-lemma identity everywhere. In one place the formal development proves the same conclusion by a different route, and we mark those rows ALT: Case P, where the paper gives a single uniform argument for the whole line (Lemma 4.15) while Lean instantiates  $r \in \{3, 4\}$  and routes the rest through Lemma 4.17 (Remark 4.16). The formal statement proved is the one the assembly consumes; it is not a weaker result. Everywhere else — including Case T, which now uses the same merge-robust route in the text and in Lean — the prose and the formalization follow the same argument.

| Result here                                                              | Lean declaration                                                                         | File                                                                                                               |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Theorem A (dichotomy)                                                    | <code>erdos_1112</code>                                                                  | <code>Final.lean</code>                                                                                            |
| Theorem 2.1                                                              | <code>erdos_1112_existence_bound,</code><br><code>existence_bound</code>                 | <code>Final.lean,</code><br><code>Existence/Nested.lean</code>                                                     |
| Strong non-existence                                                     | <code>erdos_1112_strong_nonexistence</code>                                              | <code>Final.lean</code>                                                                                            |
| Theorem A, integer form<br>( $\mathbb{Z} \rightarrow \mathbb{N}$ bridge) | <code>erdos_1112_int,</code><br><code>question_iff_questionInt</code>                    | <code>Final.lean,</code><br><code>Erdos1112.lean</code>                                                            |
| Lemma 2.2 (free gap)                                                     | <code>exists_safe_subinterval</code>                                                     | <code>Existence/FreeGap.lean</code>                                                                                |
| Lemma 3.1 (certificate)                                                  | <code>cert_hit</code>                                                                    | <code>NonEx/Certificate.lean</code>                                                                                |
| Lemmas 3.3, 3.5                                                          | <code>tailCoveringN_of_single_letter,</code><br><code>..._of_eventually_periodic</code>  | <code>NonEx/GapWord.lean</code>                                                                                    |
| Lemma 3.6 (interval)                                                     | <code>Wset_interval</code>                                                               | <code>NonEx/TwoLetter/Core.lean</code>                                                                             |
| Lemma 3.7 (sweep)                                                        | <code>sweep</code>                                                                       | <code>NonEx/TwoLetter/Core.lean</code>                                                                             |
| Lemma 3.8 (width dichotomy)                                              | <code>width_of_unbalanced</code>                                                         | <code>NonEx/TwoLetter/Core.lean</code>                                                                             |
| Prop. 3.9 (balanced)                                                     | <code>balanced_classification</code><br><code>disc_osc_le_one, mechanical_tail</code>    | <code>NonEx/TwoLetter/Balanced.lean</code><br><code>NonEx/TwoLetter/MH/</code><br><code>IrrationalCase.lean</code> |
| Lemma 3.10 (boundary)                                                    | <code>slope, slope_le_one</code>                                                         | <code>NonEx/TwoLetter/MH/Slope.lean</code>                                                                         |
| Lemma 3.11 (Sturmian)                                                    | <code>width_even_boundary</code>                                                         | <code>NonEx/TwoLetter/Core.lean</code>                                                                             |
|                                                                          | <code>tailCovering_of_sturmian,</code><br><code>walk_enters</code>                       | <code>NonEx/TwoLetter/Sturmian.lean</code><br><code>NonEx/TwoLetter/MH/Walk.lean</code>                            |
| Lemma 3.14 (slot)                                                        | <code>tailCovering_of_three_letters</code>                                               | <code>NonEx/SlotLemma.lean</code>                                                                                  |
| Lemma 4.2 (two-generator)                                                | <code>twoGen_interval, twoGen_hasRun</code>                                              | <code>Sharp/TwoGen.lean</code>                                                                                     |
| Lemma 4.3 (frame)                                                        | <code>frame_lemma</code>                                                                 | <code>Sharp/Frame.lean</code>                                                                                      |
| Lemma 4.4 (staircase)                                                    | <code>staircase_phase_base,</code><br><code>staircase_phase_extended</code>              | <code>Sharp/Staircase.lean</code>                                                                                  |
| Lemma 4.6 ( $\lambda$ -lift)                                             | <code>FrameCert.lift</code>                                                              | <code>Sharp/Lift.lean</code>                                                                                       |
| Lemma 4.7 (Graham)                                                       | <code>sharpAt_of_hardcore</code>                                                         | <code>Sharp/Graham.lean</code>                                                                                     |
| Lemma 4.13 (minimal alphabets)                                           | <code>sharp_of_minimal</code>                                                            | <code>Sharp/Graham.lean</code>                                                                                     |
| Lemma 4.14 (Case D)                                                      | <code>caseD</code>                                                                       | <code>Sharp/CaseD.lean</code>                                                                                      |
| Lemmas 4.15, 4.17 (Case P)                                               | <code>caseP_pair, caseP_r3,</code><br><code>ALT (see above) caseP_r4, caseP_large</code> | <code>Sharp/CaseP.lean</code>                                                                                      |
| Lemma 4.19 (Case L)                                                      | <code>caseL</code>                                                                       | <code>Sharp/CaseL.lean</code>                                                                                      |
| Lemma 4.20 (Case E)                                                      | <code>caseE</code>                                                                       | <code>Sharp/CaseE.lean</code>                                                                                      |
| Lemmas 4.21, 4.23 (Case T)                                               | <code>caseT, T_tail_line, tSuppT</code>                                                  | <code>Sharp/CaseT*.lean</code>                                                                                     |
| Case B (+ descent)                                                       | <code>caseB, rowFor, caseBComplete</code>                                                | <code>Sharp/CaseB.lean,</code><br><code>Sharp/CaseBClasses.lean</code>                                             |
| Tables A/B (data)                                                        | <code>certTableA_ok, certTableB_ok</code>                                                | <code>Sharp/TablesData.lean</code>                                                                                 |
| Theorem B (bounded subset-sum)                                           | <code>sharp</code>                                                                       | <code>Sharp/Main.lean</code>                                                                                       |
| Axiom audit                                                              | <code>three #print axioms commands</code>                                                | <code>AxiomsCheck.lean</code>                                                                                      |

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